



Bioinformatics MSc
Research Project List

For Session 2025-2026

Making your choice

Research projects are allocated using an automated optimisation algorithm designed to be fair and transparent. Each student submits a first, second, and third project choice, along with a list of ten additional projects they would be happy to work on. The algorithm assigns a *disappointment score* to each possible student–project pairing, with no disappointment for first choices and increasing disappointment for lower-ranked options.

Supervisor workload is also taken into account. Some supervisors offer multiple projects, and supervising additional students carries increasing penalties, encouraging a balanced distribution of students across staff where possible. This benefits both staff and students by ensuring supervision responsibilities are shared fairly.

The algorithm identifies the allocation that minimises total student disappointment plus supervisor workload penalties. As a result, student preferences are prioritised overall, but small compromises may be made to avoid large imbalances in supervision. The final allocation represents the best overall balance between student preferences and fair supervision, applied consistently across the entire cohort. While you may not receive your first choice, the system ensures outcomes are fair, reproducible, and based on explicit rules rather than subjective decisions.

Confirm your preferences using the form below.

[MSc Bioinformatics research project – Fill in form](#)

BIOINFORMATICS RESEARCH PROJECT SUPERVISOR PROFILES: 2025/2026

(See the School webpage [Our People](#) for more information on the research of individual staff; note, however, that the project areas offered below may not encompass the entire research field of the staff member concerned.)

Table of Contents

TABLE OF CONTENTS	3
1. GENOMIC BASIS OF ADAPTIVE TRAIT DIVERGENCE IN AFRICAN CICHLID FISHES	6
2. HEROES IN A HOT-SHELL: ARE COMMON WHELK ADAPTING TO WARMER OCEANS?	7
3. TRANSCRIPTOMIC CHARACTERISATION OF THE TISSUE AND CELLULAR RESPONSE DURING OCULAR INFLAMMATION	8
4. NUCLEOSOME PROFILING OF CELL-FREE DNA FROM COLORECTAL CANCER PATIENTS	9
5. THE COMPARATIVE GENOMICS OF FUNGI	10
6. THE ROLE OF DIFFERENTIAL SPLICING IN BEHAVIOURAL TRANSCRIPTOMICS	11
7. EXPLORING POTENTIAL GENETIC DETERMINANTS OF INFORMATION USE BEHAVIOUR IN HONEYBEES.....	12
8. DISSECTING GENE REGULATORY NETWORKS IN THE FLORAL MERISTEM TO CARPEL TRANSITION	13
9. ACCESSING THE EVOLUTIONARY SIGNAL IN ANCIENT PROTEINS.....	15
10. HOW SHOULD WE ESTIMATE ANCESTRAL STATES?	16
11. WHAT DOES MOLECULAR DISPARITY TELL US ABOUT EVOLUTIONARY DIVERSIFICATION?	17
12. DO ECOSYSTEMS “REMEMBER” STRESS? QUANTIFYING LEGACY EFFECTS OF EXTREME SUMMERS ON CARBON UPTAKE.....	18
13. CAN WE USE MOSQUITO TRANSCRIPTOMES TO BETTER UNDERSTAND FLORAL HUMIDITY AS A SIGNAL?	19
14. PREDATOR RETURNED - IMPACT OF RE-COLONISATION OF A LAKE DISTRICT WOODLAND BY PINE MARTENS ON LOCAL SONGBIRD POPULATIONS	21
15. COMPUTATIONAL SEQUENCE DESIGN FOR THE CONTROLLED ASSEMBLY OF DE NOVO MEMBRANE PROTEINS	23
16. PREDICTION OF NEUROPEPTIDE - RECEPTOR PAIRS IN THE NERVOUS SYSTEM OF EVOLUTIONARY ANCIENT CTENOPHORA.....	25
17. BENCHMARKING AI STRUCTURE PREDICTION TOOLS FOR TCR-PHLA DRUG DESIGN	26
18. COMPUTATIONAL EVALUATION AND REDESIGN OF DIELS-ALDER ENZYMES FOR USE AS BIOCATALYSTS IN DRUG SYNTHESIS	28
19. BIOINFORMATIC ANALYSIS OF CIRCADIAN RHYTHMS IN PERIPHERAL SENSORY NEURONS	30
20. REVERSE ENGINEERING ANIMAL PROTECTIVE COLOURATION USING GENERATIVE ADVERSARIAL NETWORKS	32
21. COMPARATIVE GENOMIC ANALYSIS OF SYMBIOTIC AND APOSYMBIOTIC STATES IN MOLLUSKS	34
22. COMPARATIVE GENOMIC ANALYSIS OF SYMBIOTIC AND APOSYMBIOTIC STATES IN FLATWORMS (ACOELS)	35

23. COMPARATIVE GENOMIC ANALYSIS OF SYMBIOTIC AND APOSYMBIOTIC CNIDARIANS	36
24. THE GEOGRAPHICAL DISTRIBUTION AND ECOLOGY OF SYNECHOCOCCUS SP. PCC 7336, POTENTIALLY ONE OF THE FIRST CYANOBACTERIA TO INVAD THE SURFACE OCEAN.....	37
25. DATING THE EVOLUTION OF NUTRIENT ACQUISITION STRATEGIES IN ANCIENT BACTERIA.....	39
26. INTEGRATIVE OMICS ANALYSIS OF THE TUMOUR MICROENVIRONMENT	41
27. QUANTIFYING THE AUNTS AND UNCLES EFFECT IN PARENTAGE ANALYSIS	42
28. ESTABLISHING THE PROVENANCE OF MUSEUM SPECIMENS OF KĀKĀ AND KĀKĀPŌ.....	44
29. GENOME-WIDE MAPPING OF GENETIC VARIATION IN A WILD ANEMONEFISH POPULATION	45
30. DIFFERENTIALLY EXPRESSED GENES IN CIRCADIAN CENTRES OF THE MAMMALIAN BRAIN	46
31. AI-POWERED DESIGN OF ADDOMER NANOPARTICLE-BASED SYNTHETIC SUPERVACCINES	47
32. SPATIAL TRANSCRIPTOMIC ANALYSIS OF PLANT IMMUNE REGULATOR PDE1 (USING SINGLE- CELL RNA-SEQ)	49
33. AI-DRIVEN DISCOVERY OF BACTERIAL STRATEGIES TO MANIPULATE HOST IMMUNE SIGNALLING	50
34. MODELLING AND INVESTIGATING EVIDENCE OF NATURAL SELECTION IN HYBRIDISING POPULATIONS.....	52
35. PREDICTING BRAIN INTERCELLULAR COMMUNICATIONS PATHWAYS FROM SINGLE CELL RNASEQ DATA	53
36. EXPLORING THE ROLE OF AI-BASED STRUCTURAL PREDICTION TO IMPROVE AMR RESISTANCE PREDICTION FROM BACTERIAL GENOMES.....	55
37. INCREASING ANALYSIS THROUGHPUT FOR AN ASSAY TO MODEL KIDNEY HEALTH	57
38. DEVELOPING THE FIRST GWAS ASSOCIATED WITH VASCULAR GLYCOCALYX CHANGES IN AVON LONGITUDINAL STUDY OF CHILDREN OF THE 90'S (ALSPAC)	59
39. INCIDENCE OF REOPERATION AFTER PRIMARY HIP OR KNEE REPLACEMENT IN ENGLAND	61
40. IDENTIFYING GENE EXPRESSION SIGNATURES FOR KIDNEY DISEASE PHENOTYPES	62
41. IDENTIFYING MULTIPLE GENETIC VARIANTS ASSOCIATED WITH KIDNEY DISEASE SUBTYPES USING METACCA AND OTHER JOINT STATISTICAL TESTS	64
42. UNRAVELLING THE MOLECULAR MECHANISMS OF SKIN INFECTION USING COMPUTATIONAL APPROACHES	66
43. OPEN PROJECTS ON EVOLUTION - PISANI LAB.....	68
44. OPEN PROJECTS ON EVOLUTION, COMPARATIVE GENOMICS, MOLECULAR PHYLOGENETICS - PAPS MONTSERRAT LAB	69
45. OPEN PROJECTS ON ENGINEERING OF BIOLOGICAL SYSTEMS - GOROCHOWSKI LAB	70
46. USING PHYLOGENETICS TO INFORM AVIAN INFLUENZA EPIDEMIOLOGY AND MULTIPLEX SEROLOGICAL SURVEILLANCE	71

1. Genomic basis of adaptive trait divergence in African cichlid fishes

Dr Martin Genner

m.genner@bristol.ac.uk

General research field

Evolutionary Biology

Project description

In Lake Malawi, around 1000 species of endemic fish species have radiated from common ancestry over the last 1 million years. Our understanding of the genetic basis of this divergence is however limited. Yet, it is now possible to explore this more comprehensively using genome resequence data has been recently published from over 200 Lake Malawi cichlid fish species. This project will explore the potential for genome-wide analysis method to uncover genetic variants and associated genes that have contributed to diversification of these fishes.

Three research papers in the area

<https://doi.org/10.1126/science.adr9961>

<https://doi.org/10.1038/s41559-018-0717-x>

<https://doi.org/10.1093/molbev/msaf147>

Biological systems involved, Area of study, Other key words associated with the project

Aquatic ecosystem; Eukaryote; Animal; Genomic; Morphology; Exploratory;

Bioinformatics field, Biological data & Programming language(s) used

Genomics; Comparative genomics; DNA; R ;Bash;

Standard software for genome mapping, variant calling and GWAS will be employed primarily using the UoB HPC, with downstream visualisation/analysis in R.

Helpful previous knowledge

An interest in evolutionary biology and adaptive radiation would be useful, alongside skills in bash scripting and R.

2. Heroes in a hot-shell: Are common whelk adapting to warmer oceans?

Dr Hugo Harrison and Prof Mark Beaumont

hugo.harrison@bristol.ac.uk

General research field

Ecology and Evolution

Project description

Many marine species are being displaced by ocean warming, actively moving polewards to remain within the limits of their physiological tolerance to heat stress. However, for species that lack a dispersive phase or are limited in their movement, they must increase their thermal tolerance to match their environment in order to survive. The common whelk, *Buccinum undatum*, is one such species. Although we don't know its physiological limits, it displays unusually high isolation by distance for a marine species and is a good candidate to explore how marine species are adapting to increasing ocean temperatures. The common whelk is the fourth most valuable fishery in the UK and the largest edible marine gastropod in the North Atlantic. Its response to ocean warming will have important economic implication for UK fisheries.

In this project, you will:

- 1) Learn to catalogue SNP loci from a published double digest RAD library of 195 whelk from England and Ireland using STACKS (Catchen et al., 2013);
- 2) Quantify the relative contribution of environmental variables on the genetic structure of whelk using redundancy analyses (Capblancq and Forester 2021)
- 3) Identify ancestry informative markers to further investigate the population structure of whelk around the British Isles.

You will receive further training in working on the HPC and R to conduct all analyses, collaborate with colleagues at the Met Office to obtain environmental variables (e.g., sea bottom temperatures), and CEFAS to inform the fisheries management for the whelk fishery. Outcomes will advance our understanding of species responses to warming ocean and deliver a set of informative markers for future studies. The ideal candidate will have an interest in population genomics and evolutionary biology.

Three research papers in the area

<https://www.sciencedirect.com/science/article/pii/S0165783622001953>

<https://onlinelibrary.wiley.com/doi/abs/10.1111/mec.12354>

<https://besjournals.onlinelibrary.wiley.com/doi/full/10.1111/2041-210X.13722>

Biological systems involved, Area of study, Other key words associated with the project

Aquatic ecosystem; Climate Change; Evolution; Genomic; Ecology

Bioinformatics field, Biological data & Programming language(s) used

Population genomics; DNA; R; Bash; STACKS, R

3. Transcriptomic characterisation of the tissue and cellular response during ocular inflammation

Dr Alison Clare and Dr Dave Copeland

alison.clare@bristol.ac.uk

General research field

Ocular Immunology

Project description

Uveitis is an umbrella term for a group of heterogeneous inflammatory conditions of the eye, which can be caused by infectious agents or autoimmune conditions. Using various chronic and acute models of ocular inflammation, we apply in vivo imaging, flow cytometry, immunohistochemistry and RNA-sequencing, to study the cellular and tissue response in the eye during disease. One model in particular is a gene therapy associated uveitis (GTAU) – whereby delivery of AAV gene therapy to treat disease leads to unwanted inflammatory side effects. This response differs and can be more pronounced depending on age, sex or disease background. In this project you will characterise both the tissue-specific response, as well as infiltrating immune cells during GTAU, and explore how age or sex may impact the response. Using single cell RNA-sequencing data taken from normal mouse retina you will deconvolve our whole retina bulk-seq datasets from AAV injected eyes in old and young mice of both sex. This approach will allow us to unpick the local tissue response to AAV gene therapy across age and sex.

Three research papers in the area

[https://www.cell.com/molecular-therapy-family/molecular-therapy/fulltext/S1525-0016\(25\)00032-2](https://www.cell.com/molecular-therapy-family/molecular-therapy/fulltext/S1525-0016(25)00032-2)

<https://star-protocols.cell.com/protocols/1386>

<https://www.biorxiv.org/content/10.1101/2025.09.09.673285v1.full#sec-44>

Biological systems involved, Area of study, Other key words associated with the project

Model organism; Animal; Neuroscience; Ophthalmology; Exploratory; Medical/clinical

Bioinformatics field, Biological data & Programming language(s) used

Transcriptomics; RNA; R; Bash

4. Nucleosome profiling of cell-free DNA from colorectal cancer patients

Dr Francisca Segers and Dr Adam Chambers (School of Cellular and Molecular Medicine)

francisca.segers@bristol.ac.uk

General research field

Cancer genomics

Project title

Nucleosome profiling of cell-free DNA from colorectal cancer patients

Project description

Circulating free DNA, or cell-free DNA (cfDNA), is DNA circulating in the bloodstream after being released during programmed cell death. Cancer patients generally have elevated levels of cfDNA, and the cfDNA is then also referred to as circulating tumour DNA (ctDNA). Recently, research into ctDNA has expanded, for its promising applications in cancer diagnostics. For example, the quantity and the characteristics of the ctDNA can be markers for cancer therapy effectiveness and tumour subtyping, taking away the need for complicated tissue biopsies in patients. More specifically, fragmentation patterns of ctDNA reflect the chromatin structure of the cells from which the ctDNA originated: Nucleosomes protect the DNA from fragmentation, resulting in decreased sequencing coverage in regions of open chromatin, potentially informing us of gene activity in tumour cells. In this project, the student will test bioinformatic software (e.g LIQUORICE, Griffin) to generate nucleosome profiles from ctDNA sequencing datasets originating from patients with and without colorectal cancer. These profiles will then be examined for their association with disease presence and progression.

Three research papers in the area

Doebley, A. L., Ko, M., Liao, H., Cruikshank, A. E., Santos, K., Kikawa, C., ... & Ha, G. (2022). A framework for clinical cancer subtyping from nucleosome profiling of cell-free DNA. *Nature Communications*, 13(1), 7475.

Hallermayr, A., Wohlfrom, T., Steinke-Lange, V., Benet-Pagès, A., Scharf, F., Heitzer, E., ... & Pickl, J. M. (2022). Somatic copy number alteration and fragmentation analysis in circulating tumor DNA for cancer screening and treatment monitoring in colorectal cancer patients. *Journal of Hematology & Oncology*, 15(1), 125.

Peneder, P., Bock, C., & Tomazou, E. M. (2022). LIQUORICE: detection of epigenetic signatures in liquid biopsies based on whole-genome sequencing data. *Bioinformatics Advances*, 2(1), vbac017.

Biological systems involved, Area of study, Other key words associated with the project

Human; Medical/clinical

Bioinformatics field, Biological data & Programming language(s) used

Genomics; DNA; Bash; R; Griffin, LIQUORICE

Helpful previous knowledge

Molecular biology

5. The comparative genomics of Fungi

Dr Francisca Segers

francisca.segers@bristol.ac.uk

General research field

Comparative genomics

Project title

The comparative genomics of Fungi

Project description

The Fungal kingdom is a remarkably diverse clade of eukaryotes, encompassing a wide range of lifestyles and found in a plethora of habitats. Fungal detritivores play important roles in nutrient cycles, but fungi can also act as parasites and pathogens of many different organisms. Plant-fungal mutualistic associations, such as mycorrhizas and lichens, have put a stamp on terrestrial habitats, even the ones hostile to most life forms. Many lifestyles have evolved independently across the fungal tree of life, which provides the opportunity to explore whether similar molecular mechanisms were recurrently involved in enabling certain niche shifts. Using comparative genomic methods and data from public repositories, you can characterise the genomic changes associated with the evolutionary emergence of a fungal lifestyle of your interest, e.g., lichens, plant symbionts, animal pathogens, or species of interest to industrial biotechnology, such as those that can be used for food processing or the manufacturing of antibiotics.

Three research papers in the area

Hess, J., Skrede, I., Chaib De Mares, M., Hainaut, M., Henrissat, B., & Pringle, A. (2018). Rapid divergence of genome architectures following the origin of an ectomycorrhizal symbiosis in the genus *Amanita*. *Molecular biology and evolution*, 35(11), 2786-2804.

Naranjo-Ortiz, M. A., & Gabaldón, T. (2019). Fungal evolution: major ecological adaptations and evolutionary transitions. *Biological Reviews*, 94(4), 1443-1476.

Resl, P., Bujold, A. R., Tagirdzhanova, G., Meidl, P., Freire Rallo, S., Kono, M., ... & Spribille, T. (2022). Large differences in carbohydrate degradation and transport potential among lichen fungal symbionts. *Nature communications*, 13(1), 2634.

Biological systems involved, Area of study, Other key words associated with the project

Fungi; Evolution; Genomic

Bioinformatics field, Biological data & Programming language(s) used

Comparative genomics; DNA; R; Bash; Python

6. The role of differential splicing in behavioural transcriptomics

Dr Francisca Segers

francisca.segers@bristol.ac.uk

General research field

Behavioural genomics

Project description

For many organisms the genetic basis of behaviour is still barely understood. In the last decade, the introduction of highthroughput transcriptome sequencing enabled researchers to identify genes of which the degree of expression is associated with certain behavioural phenotypes. Behavioural variation associated with one of the most complex animal communication behaviours, the honey bee waggle dance, has previously been explored on a molecular level. Genes have been found that may underpin the propensity of individuals to show interest in the information relayed by the honeybee waggle dance. The aim of this project is to dive deeper in the transcriptomic information at hand, and to assess whether differentially spliced genes are linked to observed behavioural variation in dance following among honey bees.

Three research papers in the area

Mehmood, A., Laiho, A., Venäläinen, M. S., McGlinchey, A. J., Wang, N., & Elo, L. L. (2020). Systematic evaluation of differential splicing tools for RNA-seq studies. *Briefings in bioinformatics*, 21(6), 2052-2065.

Kennedy, A., Peng, T., Glaser, S. M., Linn, M., Foitzik, S., & Grüter, C. (2021). Use of waggle dance information in honey bees is linked to gene expression in the antennae, but not in the brain. *Molecular Ecology*, 30(11), 2676-2688.

Wang, Y., Xie, Z., Kutschera, E., Adams, J. I., Kadash-Edmondson, K. E., & Xing, Y. (2024). rMATS-turbo: an efficient and flexible computational tool for alternative splicing analysis of large-scale RNA-seq data. *Nature Protocols*, 19(4), 1083-1104.

Biological systems involved, Area of study, Other key words associated with the project

Animal; Animal behaviour

Bioinformatics field, Biological data & Programming language(s) used

Transcriptomics ; RNA ; R; Bash ; rMats-Turbo

7. Exploring potential genetic determinants of information use behaviour in honeybees

Dr Francisca Segers

francisca.segers@bristol.ac.uk

General research field

Behavioural genomics

Project description

Intraspecific information transmission is vital for social organisms, yet decisions on how to utilise this information are complex, shaped by environmental and intrinsic factors. Our research focusses on the honeybee (*Apis mellifera*) waggle dance, a communication method through which foragers convey complex information about food locations to nestmates. Foragers may choose to follow information relayed by these dances to obtain vector information about the locations of food sources (social information) or rely on their own memories of previous foraging routes (private information). The molecular mechanisms guiding this choice are as yet unknown. This project will use previously published RNA-seq data for variant calling, aiming to identify SNPs associated with social or private information use. We hypothesise that Single Nucleotide Polymorphisms (SNPs) differing between social and private information users affect genes involved in sensory perception and information processing. This project will involve learning how to use variant calling software, a methodology which is applicable to many research fields in biology and medicine.

Three research papers in the area

Brouard, J. S., & Bissonnette, N. (2012). Variant calling from RNA-seq data using the GATK joint genotyping workflow. In *Variant calling: Methods and protocols* (pp. 205-233). New York, NY: Springer US.

Kennedy, A., Peng, T., Glaser, S. M., Linn, M., Foitzik, S., & Grüter, C. (2021). Use of waggle dance information in honey bees is linked to gene expression in the antennae, but not in the brain. *Molecular Ecology*, 30(11), 2676-2688.

Teixeira, E. W., de Paiva Daibert, R. M., Júnior, L. A. G., da Silva, M. V. G. B., Alves, M. L. T. M. F., Evans, J. D., & Toth, A. L. (2021). Transcriptomic analysis suggests candidate genes for hygienic behavior in African-derived *Apis mellifera* honeybees. *Apidologie*, 52(2), 447-462.

Biological systems involved, Area of study, Other key words associated with the project

Animal; Animal behaviour

Bioinformatics field, Biological data & Programming language(s) used

Transcriptomics; Genomics; RNA; R; Bash; GATK, bcftools

8. Dissecting gene regulatory networks in the floral meristem to carpel transition

Dr Beatriz Goncalves

beatriz.goncalves@bristol.ac.uk

General research field

Plant developmental biology

Project description

We are interested in the evolution of flowering plants and the genes that underpin their amazing diversity of shapes. In my lab we study a floral organ called the carpel which is responsible for the production of fruit and seed. Broadly we want to understand how this structure originated, what genes control its formation, and how changes to its developmental program have led to the evolution of a diversity of fruit shapes. As carpels are a key trait involved in flowering plant reproduction, we are also interested in what traits in the carpel have contributed to the success and diversification of the most evolutionarily successful of all plant groups.

There is some flexibility in what the project will involve but it will align with ongoing work in the team investigating carpel morphogenesis and the acquisition of new cell identities during carpel development. In practice this project will build on existing knowledge of developmental pathways and genes to identify the gene regulatory network that controls carpel polarity. You will query existing transcriptomics datasets, including single cell and spatial transcriptomics as well as gene expression databases, and contrast different developmental stages, tissues, and genetic lines, to spatially and temporally resolve the genetic networks that control cell identities in the carpel. This work will be based on the model species *Arabidopsis* for which most of the transcriptomics data is available, but there is an option to extend it to other species if any relevant datasets are available adding the evolutionary dimension to the project.

Three research papers in the area

Chronology of transcriptome and proteome expression during early *Arabidopsis* flower development <https://academic.oup.com/jxb/article/76/10/2743/7945333>

A 3D gene expression atlas of the floral meristem based on spatial reconstruction of single nucleus RNA sequencing data <https://www.nature.com/articles/s41467-022-30177-y>

A single-cell, spatial transcriptomic atlas of the *Arabidopsis* life cycle <https://www.nature.com/articles/s41477-025-02072-z>

Biological systems involved, Area of study, Other key words associated with the project

Plants; Model organism; Development; Morphology

Bioinformatics field, Biological data & Programming language(s) used

Transcriptomics; Database searching; Gene families; RNA; Python; R

Helpful previous knowledge

It will help if you have some understanding of developmental biology and pathways or a keen interest in learning about plant developmental genetics. In practice RNAseq data analysis will be helpful.

9. Accessing the evolutionary signal in ancient proteins

Dr Joseph Keating

2 projects available (students will work on different matrix proteins)

joe.keating@bristol.ac.uk

General research field

Evolutionary Biology

Project description

This research area focuses on the use of ancient proteins as a source of phylogenetic information, with a particular emphasis on bone and enamel matrix proteins. Projects may investigate how protein sequences can inform evolutionary relationships, for example by assembling vertebrate protein datasets from online resources and analysing them within a supertree framework. There is also potential for projects that examine empirical datasets from extinct taxa to explore their phylogenetic placement using ancient protein data. More broadly, projects in this area aim to assess the strengths and limitations of ancient proteomics as a tool for studying evolutionary history.

Three research papers in the area

1. Baker & Buckley 2025 <https://onlinelibrary.wiley.com/doi/full/10.1111/1755-0998.14072>
2. Harvey et al 2021
<https://royalsocietypublishing.org/rsos/article/8/8/201955/96187/Phylogenetic-analyses-of-ray-finned-fishes>
3. Paterson et al 2025 <https://www.nature.com/articles/s41586-025-09231-4>

Biological systems involved, Area of study, Other key words associated with the project

Eukaryote;Animal;Fungi;fossils; Evolution;Morphology;Palaeobiology

Bioinformatics field, Biological data & Programming language(s) used

Phylogenetics/phylogenomics;Database searching;Proteomics;Genomics;
DNA;RNA;Proteins;morphology; Python;R;Bash

IQtree or equivalent phylogenetic software

Helpful previous knowledge

Interest in phylogeny and phylogenetic comparative methods.

Competent in coding in R or Python

10. How should we estimate ancestral states?

Dr Joseph Keating

joe.keating@bristol.ac.uk

General research field

Evolutionary Biology

Project description

This research area focusses on the use of ancestral state estimation in evolutionary biology. Projects may address methodological questions, such as comparing the performance of existing ancestral state estimation models using simulated datasets or developing new models. Alternatively, or in combination, projects may apply ancestral state estimation to empirical questions in early animal evolution, including the origins of key traits. The emphasis will be on understanding how methodological choices influence evolutionary inference.

1. Cockx et al 2025 <https://academic.oup.com/sysbio/advance-article/doi/10.1093/sysbio/syaf063/8253600>
2. Keating 2023 <https://www.biorxiv.org/content/10.1101/2023.08.31.555762v1.full>
3. Jiang et al 2023 <https://www.nature.com/articles/s41559-023-02074-0>

Biological systems involved, Area of study, Other key words associated with the project

Eukaryote;Animal;Fungi;fossils; Evolution;Morphology;Palaeobiology

Bioinformatics field, Biological data & Programming language(s) used

Phylogenetics/phylogenomics;Database searching;Proteomics;Genomics;
DNA;RNA;Proteins;morphology; Python;R;Bash

IQtree or equivalent phylogenetic software

Helpful previous knowledge

Interest in phylogeny and phylogenetic comparative methods.
Competent in coding in R or Python

11. What does molecular disparity tell us about evolutionary diversification?

Dr Joseph Keating

joe.keating@bristol.ac.uk

General research field

Evolutionary Biology

Project description

This research area explores patterns of molecular disparity and how they relate to morphological and functional diversity through evolutionary time. Projects may investigate questions such as how molecular disparity corresponds to morphological disparity, how different types of molecular data (e.g., genes, proteins, introns, or regulatory RNAs) compare in their patterns of disparity, or how molecular variation relates to functional diversity within large clades. Using comparative datasets and existing phylogenies, projects in this area aim to evaluate the extent to which molecular data capture different aspects of evolutionary diversification and to assess the utility of molecular disparity as a macroevolutionary measure.

Three research papers in the area

1. Van den Ende et al 2023 <https://besjournals.onlinelibrary.wiley.com/doi/full/10.1111/2041-210X.14166>
2. Keating et al 2023 <https://link.springer.com/article/10.1186/s12862-023-02131-z>
3. Smith & Donoghue 2022 <https://www.nature.com/articles/s41559-022-01844-6>

Biological systems involved, Area of study, Other key words associated with the project

Eukaryote;Animal;Fungi;fossils; Evolution;Morphology;Palaeobiology

Bioinformatics field, Biological data & Programming language(s) used

Phylogenetics/phylogenomics;Database searching;Proteomics;Genomics;
DNA;RNA;Proteins;morphology; Python;R;Bash

IQtree or equivalent phylogenetic software

Helpful previous knowledge

Interest in phylogeny and phylogenetic comparative methods.
Competent in coding in R or Python

12. Do ecosystems “remember” stress? Quantifying legacy effects of extreme summers on carbon uptake

Dr Martin De Kauwe

xj21307@bristol.ac.uk

General research field

Ecology and environmental change/climate change

Project description

Climate change is altering the frequency, intensity, and duration of extreme events such as droughts and heatwaves. Plant responses to these extremes are likely to depend not only on current environmental conditions, but also on prior exposure to stress, both within and across growing seasons. In this project, you will work with a rich global dataset of ecosystem-scale (~km scale) carbon flux measurements that capture how plant communities respond to climate variability in real time. You will apply explainable machine learning approaches to examine how carbon uptake responds to extreme events under different environmental conditions, and how these responses change when similar events occur repeatedly. The analysis will focus on identifying memory and legacy effects that are likely to be critical for improving predictions of plant resilience and recovery from summer climate extremes. You will also have the opportunity to contrast these data-driven results with simulations from process-based ecosystem models.

Three research papers in the area

- Hammond et al. Global field observations of tree die-off reveal hotter-drought fingerprint for Earth’s forests. Nat Commun 13, 1761 (2022). <https://doi.org/10.1038/s41467-022-29289-2>
- Pichler, M., & Hartig, F. (2023). Machine learning and deep learning—A review for ecologists. Methods in Ecology and Evolution, 14, 994–1016. <https://doi.org/10.1111/2041-210X.14061>
- Ezhova, E., Laanti, T., Lintunen, A., Kolari, P., Nieminen, T., Mammarella, I., Heljanko, K., and Kulmala, M.: Explainable machine learning for modeling of net ecosystem exchange in boreal forests, Biogeosciences, 22, 257–288, <https://doi.org/10.5194/bg-22-257-2025>, 2025.

Biological systems involved, Area of study, Other key words associated with the project

Plants;Terrestrial ecosystem

Climate Change;Ecology

Bioinformatics field, Biological data & Programming language(s) used

Machine Learning/AI;Data Modeling

Ecosystem flux measurements (e.g., CO₂, H₂O, energy exchange)

Python;R

Helpful previous knowledge

Some background in biology or environmental science would be helpful, but students from other quantitative or physical science backgrounds are also welcome.

13. Can we use Mosquito transcriptomes to better understand floral humidity as a signal?

Dr Heather Whitney and Prof Davide Pisani

bzyhmw@bristol.ac.uk

General research field

Pollinating insect sensory ecology

Project description

Recent work has identified new ways in which pollinators can detect rewarding flowers. One of these new modalities is floral humidity, where flowers have distinct regions of high humidity. Several studies have found that a range of pollinators are able to detect/learn/have an innate preference for these floral humidity patterns, but it is still unclear to what extent these humidity patterns are an important signal that contributes to pollination efficiency.

One way to work out how important humidity is as a potential pollination signal is to look at species that use flowers as a food source only some of the time. Mosquitos are an important pollinator, with both the male and female mosquitoes initially obtaining their energy from nectar. However, female mosquitoes then change their target and use olfaction, temperature and humidity sensors to find blood meals, before finally searching for bodies of water for egg ovipositing. The genes for mosquito humidity sensors have been identified. There are transcriptomes of this change, and we would like to use these transcriptomes to answer questions potentially including:

- i) Are female humidity sensors expressed differently at different stages (when searching for nectar/blood meal/water for oviposition)
- ii) Are male and female humidity sensors expressed differently?
- iii) Can we compare the expression of humidity sensors with specific volatile receptors?
- iv) Can we compare mosquito humidity sensors with the genomes/transcriptomes of other insect pollinators?
- v) Can we look at the evolution of insect humidity sensors and see if there is any change potentially driven by flower/pollinator co-evolution?

Three research papers in the area

Harrap, M.J., Hempel de Ibarra, N., Knowles, H.D., Whitney, H.M. and Rands, S.A., 2020. Floral humidity in flowering plants: a preliminary survey. *Frontiers in Plant Science*, 11, p.249.

Laursen, W.J., Budelli, G., Tang, R., Chang, E.C., Busby, R., Shankar, S., Gerber, R., Greppi, C., Albuquerque, R. and Garrity, P.A., 2023. Humidity sensors that alert mosquitoes to nearby hosts and egg-laying sites. *Neuron*, 111(6), pp.874-887.

de Carvalho, S.S., Rodovalho, C.M., Gaviraghi, A., Mota, M.B.S., Jablonka, W., Rocha-Santos, C., Nunes, R.D., Sa-Guimaraes, T.D.E., Oliveira, D.S., Melo, A.C. and Moreira, M.F., 2021. *Aedes aegypti* post-emergence transcriptome: Unveiling the molecular basis for the hematophagic and gonotrophic capacitation. *PLoS Neglected Tropical Diseases*, 15(1), p.e0008915.

Biological systems involved, Area of study, Other key words associated with the project

Plants;Animal; Agriculture;Evolution;Ecology;Animal behaviour;pollination
pollination, flowers, insects, signals, co-evolution

Bioinformatics field, Biological data & Programming language(s) used

Transcriptomics

Helpful previous knowledge

Understanding of some aspect of sensory ecology or pollination biology

14. Predator returned - impact of re-colonisation of a Lake district woodland by Pine martens on local songbird populations

Prof Marc Holderied

marc.holderied@bristol.ac.uk

Other research staff, PhD students or technicians involved (if applicable)

Will Cresswell, recent PhD graduate

General research field

Automatic Passive Acoustic Wildlife Monitoring

Project title

Predator returned - impact of re-colonisation of a Lake district woodland by Pine martens on local songbird populations

Project description

Based on an existing dataset of millions of acoustic recordings (and camera trap footage) gathered over the last three years, which amazingly spans and documents the arrival and re-establishment of a main UK predator, the Pine marten, we will use the online Arbimon platform and machine learning approaches to quantify the effect their arrival had on vocal activity in a wide range of vocally active species. We already know that invasive Grey squirrels decline and native Red squirrel numbers soar (PhD project of Dr Cresswell, graduating 2026), and pilot analyses on six songbird species shows a massive decline in vocal activity by up to 95% in one bird species. Using this unique dataset, we are looking for a student with the skills and drive to work with and query a very large dataset with complex spatiotemporal data structures (including deep learning approaches for classifiers) to draw out the actual ecological effect of a momentous faunal change that turns the tide for the native Red squirrels against their invasive Grey squirrel competitors that have replaced them throughout England.

Three research papers in the area

For this project:

(1) Cresswell, Baker & Holderied (submitted). Pine martens tip the scales: the biotic and abiotic factors that shape coexistence of native and invasive squirrel species. - Ask Marc for a copy if not yet published.

For Arbimon platform see the many examples here <https://arbimon.org/publications> e.g.

(2) Ramesh, Hariharan, Akshay, Choksi, Khanwilkar, DeFries, Robin, Using passive acoustic monitoring to examine the impacts of ecological restoration on faunal biodiversity in the Western Ghats, Biological Conservation, Volume 282, 2023, 110071, ISSN 0006-3207, <https://doi.org/10.1016/j.biocon.2023.110071>

(<https://www.sciencedirect.com/science/article/pii/S0006320723001726>)

(3) Müller, J., Mitesser, O., Schaefer, H.M. et al. Soundscapes and deep learning enable tracking biodiversity recovery in tropical forests. Nat Commun 14, 6191 (2023).

<https://doi.org/10.1038/s41467-023-41693-w>

Biological systems involved, Area of study, Other key words associated with the project

Eukaryote;Terrestrial ecosystem;Animal; Ecology;Climate Change;Sustainability;Rewilding

Bioinformatics field, Biological data & Programming language(s) used

Machine Learning/AI;Database searching; Images/video;Acoustic recordings; Arbimon;Python;R;
Arbimon online platform; Perhaps Avisoft SASLAB pro for acoustic template preparation

15. Computational sequence design for the controlled assembly of de novo membrane proteins

Dr Paul Curnow and Prof Ross Anderson

p.curnow@bristol.ac.uk

Other research staff, PhD students or technicians involved (if applicable)

Ben Hardy (PDRA), Harrison Phillips (MScR)

General research field

Computational protein design

Project title

Computational sequence design for the controlled assembly of de novo membrane proteins

Project description

We have recently described a first-in-class computationally-designed integral membrane protein. This protein – which we call ‘CytbX’ - is a new model system for the design and assembly of heme-binding bioenergetic membrane proteins. We are now interested in pursuing the redesign of CytbX to enable the controlled lateral assembly of this protein in the membrane plane. This initially includes specific and symmetrical oligomeric states – dimer, trimer, tetramer, pentamer, hexamer. The concept is that these controlled assemblies can then be developed into higher-order protein ‘sheets’ similar to the crystalline hexagonal lattice seen in bioenergetic purple membranes and semicrystalline arrays of photosynthetic membrane proteins. This includes adding fusion protein partners that extend the functionality of the sheets beyond the membrane plane. We envisage the ultimate application of these nanoscale conductive protein sheets as novel synthetic bioelectronic and photoactive soft materials. The student will implement and develop our existing design pipeline, using tools such as MPNN on ISAMBARD-AI.

Three research papers in the area

Hardy, B.J., A. Martin Hermosilla, D.K. Chinthapalli, C.V. Robinson, J.L.R. Anderson, and P. Curnow, Cellular production of a de novo membrane cytochrome. Proc Natl Acad Sci U S A, 2023. 120(16): p. e2300137120.

Hardy BJ and Curnow P (2024) Computational design of de novo bioenergetic membrane proteins. Biochem Soc. Trans. 52 (4): 1737-45

Biological systems involved, Area of study, Other key words associated with the project

Molecular systems – proteins; Exploratory; Methods; Biochemistry; Synthetic Biology; Structural Biology

Bioinformatics field, Biological data & Programming language(s) used

Machine Learning/AI; Bioengineering; Protein design; Python; Bash; MPNN for protein sequence design; AlphaFold or similar (e.g. Boltz)

Helpful previous knowledge

Some knowledge of protein science might be helpful but is not an absolute requirement.

Other relevant information

This is part of an ongoing project and the student is not expected to set things up from scratch! My lab combines experimental and computational work and the student will not be out on a limb.

16. Prediction of neuropeptide - receptor pairs in the nervous system of evolutionary ancient Ctenophora

Maria Sachkova and Dr Pierre Buscaill

2 projects available (students will work on different neuropeptides)

maria.sachkova@bristol.ac.uk

Other research staff, PhD students of technicians involved (if applicable)

Aidan Pugh

General research field

Evolutionary neuroscience

Project title

Prediction of neuropeptide - receptor pairs in the nervous system of evolutionary ancient Ctenophora

Project description

Ctenophora (or comb jellies) are marine invertebrates representing an evolutionary ancient lineage of animals, that was first to evolve nervous system. Ctenophore nervous system is very poorly characterized and has multiple unique features. It has been even hypothesized to have evolved independently from the rest of animals. Recently, many unique neuropeptides have been identified in ctenophore neurons; however it is still unknown what receptors they target. In this project, we will attempt to predict receptors for the ctenophore neuropeptides using AI-based AlphaFold tool to model peptide-receptor complexes.

Three research papers in the area

<https://www.sciencedirect.com/science/article/pii/S2405471224003016?via%3Dihub>

<https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/pmic.202300397>

<https://www.sciencedirect.com/science/article/pii/S0960982221012458>

Biological systems involved, Area of study, Other key words associated with the project

Eukaryote; Animal; Evolution; Methods; Neuroscience; Exploratory; Structural Biology

Bioinformatics field, Biological data & Programming language(s) used

Proteomics; Machine Learning/AI; Proteins; Python

AlphaFold (<https://alphafoldserver.com/>,

<https://colab.research.google.com/github/sokrypton/ColabFold/blob/main/AlphaFold2.ipynb>)

Helpful previous knowledge

Protein structure, some ideas about neuropeptides and membrane receptors.

Going through this online course will be helpful:

<https://www.ebi.ac.uk/training/online/courses/alphafold/>

17. Benchmarking AI Structure Prediction Tools for TCR-pHLA Drug Design

Dr Marc Van der Kamp

marc.vanderkamp@bristol.ac.uk

Other research staff, PhD students or technicians involved (if applicable)

Tania Gardasevic

General research field

Structural bioinformatics

Project description

The interaction between T-cell receptors (TCR) and peptide-human leukocyte antigen (pHLA) molecules enables 'effector' T-cells to initiate killing of aberrant cells. The interaction is vital to the clearance of pathogens and plays a major role in autoimmunity and transplantation. Through designing soluble TCRs that can be fused with immune activating antibody fragments, bispecific T-cell engagers have been developed that can target key cancer antigens (ImmTACs®, Immune mobilising monoclonal T-cell receptors Against Cancer).

To achieve desired soluble TCRs suitable for treatment of cancers as well as infectious and autoimmune diseases, both affinity and specificity of TCR-pHLA binding need to be optimised. Natural TCRs recognize pHLA with micromolar affinities, and this affinity needs to be enhanced several thousand fold towards their antigen (whilst retaining or enhancing selectivity to specific target peptides). Optimising TCRs is thus challenging and currently requires extensive rounds of lab-based generation and evaluation. Computational modelling and design has the potential to significantly reduce the time and resources required, but such approaches come with their own challenges. For example, standard AI-based methods for protein structure prediction, although now often highly accurate for most proteins, tend to perform quite poorly for the prediction of the hypervariable TCR loops that directly interact with the pHLA. However, accurately predicting TCR-pHLA structures is an essential step towards accurate affinity estimation, and thus for computational design of affinity-enhanced TCRs for therapeutic use.

Recently, we evaluated the accuracy of a number of AI-based structure prediction tools by comparison to experimentally determined TCR-pHLA complex structures of low, medium and high binding affinity. This has revealed promising ways forward, in particular if structural information on homologous TCR-pHLA complexes is available. In this project, the aim is to explore this further by examining the following questions: 1) Are the structure prediction protocols transferable to other TCR-pHLA complexes? 2) Can the resulting complex structures be directly used to rank affinities of TCR variants? For the latter, physics-based and AI-based scoring functions will be evaluated. If successful, the project will contribute directly to the development of a computational workflow for TCR affinity engineering, which in turn will speed up the development of therapeutic TCR-based biologic drugs for the treatment of cancer as well as infectious and autoimmune diseases.

Three research papers in the area

Crean et al. (2020) Molecular Rules Underpinning Enhanced Affinity Binding of Human T Cell Receptors Engineered for Immunotherapy, Mol. Ther. Oncolytics 18, 443-456.
<https://doi.org/10.1016/j.omto.2020.07.008>

Poole et al. (2022) Therapeutic high affinity T cell receptor targeting a KRASG12D cancer neoantigen. Nat. Commun. 13, 5333. <https://www.nature.com/articles/s41467-022-32811-1>

Quast, N.P., Abanades, B., Guloglu, B. et al. T-cell receptor structures and predictive models reveal comparable alpha and beta chain structural diversity despite differing genetic complexity. Commun Biol 8, 362 (2025). <https://doi.org/10.1038/s42003-025-07708-6>

Biological systems involved, Area of study, Other key words associated with the project

Structural Biology; Biochemistry; Treatment/Therapeutics; Machine Learning/AI; Bioengineering; AI-based structure prediction, biologic drug design

Bioinformatics field, Biological data & Programming language(s) used

Proteins; Python; AI structure prediction software, possibly Amber for MD simulation (ambermd.org)

Helpful previous knowledge

Biochemistry, Protein structure visualisation and analysis, AI-based structure prediction

Other relevant information

Tania's poster related to this project was awarded a prize at a recent international conference. You can read about it and view the poster here:

<https://www.embl.org/about/info/course-and-conference-office/2026/01/meet-the-poster-prize-winners-of-computational-structural-biology/>

18. Computational evaluation and redesign of Diels-Alder enzymes for use as biocatalysts in drug synthesis

Dr Marc Van der Kamp

marc.vanderkamp@bristol.ac.uk

Other research staff, PhD students of technicians involved (if applicable)

Magdalena Kazmierczak

General research field

Structural bioinformatics, computational enzymology

Project description

Enzymes are remarkable biocatalysts that allow rapid, (stereo)selective and efficient catalysis under mild conditions. Biotechnology exploits the rate-enhancing properties of these natural biocatalysts to manufacture high added-value products. Of particular current interest are Diels-Alderase, enzymes that catalyse the Diels-Alder reaction, a reaction widely recognised as one of the cornerstone synthetic organic reactions of the 20th century. Diels-Alderase involved in making spirotetronate antibiotics are typically small and stable beta-barrel proteins, which makes them attractive as industrial biocatalysts. Based on sequence and structural similarity, many such putative Diels-Alderase have been found, even without knowing the substrate(s) that these enzymes act on in nature. For specific enzyme-substrate combinations, we have previously used and optimized computational protocols for docking, reaction simulation and affinity prediction that previously for the highly stable Diels-Alderase AbyU and a few homologues of particular interest (AbmU, Cyc15).

To better understand which Diels-Alderase will be most suitable for a particular substrate and desired product outcome, their structures and likely complexes with specific substrate/product combinations need to be investigated. Previously, we have performed structure predictions of the catalytically-relevant 'closed' enzyme conformation for many putative beta-barrel Diels-Alderase. In this project, the aims are to analyse these structures in detail, select promising candidates for the desired product outcomes (including through structure-based engineering), and perform our molecular docking and simulation protocols to evaluate these candidates. If this evaluation indicates that the candidate Diels-Alderase are promising, they will be tested experimentally by our academic and industrial collaborators. Ultimately, this can then lead to the use of these enzymes in the sustainable production of drug molecules in industry.

Three research papers in the area

1. <https://chemistry-europe.onlinelibrary.wiley.com/doi/10.1002/cbic.202300382>
2. <https://pubs.rsc.org/en/content/articlelanding/2024/sc/d4sc03253e>
3. <https://www.sciencedirect.com/science/article/pii/S1367593116301508>

Biological systems involved, Area of study, Other key words associated with the project

Prokaryote; Sustainability; Structural Biology; Biochemistry; Synthetic Biology; molecular docking, biomolecular simulation, computational enzymology, industrial biotechnology

Bioinformatics field, Biological data & Programming language(s) used

Machine Learning/AI; Structural bioinformatics; Bioengineering; Proteins; Python; Bash; AutoDock Vina, Amber (ambermd.org)

Helpful previous knowledge

Biochemistry, protein structure visualisation, Knowledge of molecular structure / chemistry (atomic structure of amino acids, stereochemistry etc.)

19. Bioinformatic Analysis of Circadian Rhythms in Peripheral Sensory Neurons

Dr Lukasz Chrobok and Prof. Alan Champneys

lukasz.chrobok@bristol.ac.uk

Other research staff, PhD students or technicians involved (if applicable)

Emily Birt - PhD Student

General research field

Neuroscience - advanced data analysis and modelling

Project description

Life on Earth is shaped by predictable day–night cycles, and most organisms have evolved endogenous ~24-hour timekeeping systems known as circadian clocks to anticipate and adapt to these environmental changes. In mammals, the suprachiasmatic nuclei (SCN) of the hypothalamus are classically described as the primary circadian pacemaker. However, it is now well established that circadian clocks are not confined to the brain. Instead, rhythmic clock gene expression is distributed across many peripheral tissues, including components of the peripheral nervous system, where local oscillators contribute to tissue-specific physiology and behaviour.

Emerging evidence suggests that peripheral clocks can regulate sensory processing, pain sensitivity, and inflammatory responses, raising the possibility that circadian oscillators in sensory ganglia play a role in daily rhythms of nociception. The dorsal root ganglia (DRG), which house the primary sensory neurons conveying peripheral signals to the central nervous system, represent a compelling but underexplored site of circadian regulation. Understanding how circadian rhythms are organised and modulated in this peripheral oscillator requires not only high-quality experimental data, but also robust and appropriate analytical approaches.

This Master's project is embedded within an ongoing experimental programme using PER2::LUC reporter mice, which enable real-time bioluminescence monitoring of clock gene expression *ex vivo*. The student will work with long-term circadian datasets obtained from DRG tissue, typically consisting of a week-long baseline recording followed by a defined experimental intervention (pharmacological perturbation). These datasets provide an excellent opportunity to interrogate both stable rhythmicity and dynamic changes in circadian parameters such as period, amplitude, and phase.

The primary aim of the project is to systematically compare different analytical methods for circadian rhythm detection and characterisation in this peripheral sensory oscillator. The student will focus on wavelet-based analysis, implemented using the Python package PyBOAT, and will benchmark its performance against alternative commonly used approaches of their choice. Particular emphasis will be placed on evaluating how well each method captures rhythm stability and transient changes following experimental intervention.

Through this work, the student will gain hands-on experience in time-series analysis, circadian bioinformatics, and Python-based data processing, while contributing to an important methodological question: which analytical framework is best suited for detecting and quantifying

circadian rhythms in peripheral neural tissues under dynamic conditions. The project will culminate in a written report structured as a short scientific manuscript, encouraging the student to critically interpret both biological and computational findings.

Three research papers in the area

10.1016/j.neubiorev.2021.08.004; 10.1007/978-1-0716-2249-0_3; 10.1101/2025.04.07.646998

Biological systems involved, Area of study, Other key words associated with the project

Animal; Neuroscience; neuroscience, peripheral nervous system, circadian clock, time-series analysis, wavelet

Bioinformatics field, Biological data & Programming language(s) used

Data Modeling; Images/video; Continuous datasets; Python; R; Excel, Python, R, GraphPad, Actimetrics (all available in the lab)

Helpful previous knowledge

use of Python and PyBOAT, basic knowledge of circadian neuroscience

20. Reverse engineering animal protective colouration using Generative Adversarial Networks

Dr Laszlo Talas and Dr Jasmina Stevanov (Business School)

l.talas@bristol.ac.uk

Other research staff, PhD students of technicians involved (if applicable)

George Lewis (PhD student starting 2nd Feb on snake camouflage)

General research field

Visual ecology

Project description

Animals use camouflage, a form of protective colouration, to avoid detection from predators and prey. However, the effectiveness of camouflage is background dependent and often a function of background complexity. In this project, we are interested to implement Generative Adversarial Networks (GANs), a type of deep neural network architecture, to assess how camouflage develops as backgrounds become more diverse.

GANs provide a powerful framework for modelling evolutionary arms races by placing two agents in direct competition. In our case, the discriminator plays the role of a predator, learning to distinguish empty backgrounds from those containing a target, while the generator acts as prey, evolving colour and texture patterns that maximise false-negative errors (i.e., causing the predator to misclassify scenes containing a target as empty).

The generator is driven by a latent vector that can be interpreted as an artificial genotype, allowing exploration of a high-dimensional parameter space that determines expressed camouflage patterns (the artificial phenotype). Our previous work has shown that such systems can produce highly effective camouflage (e.g., Talas et al. 2020), but the relationship between latent variables and specific visual traits remains poorly understood.

The proposed MSc project will extend an existing CamoGAN framework by introducing reversibility between genotype and phenotype, for example through the incorporation of CycleGAN (Zhu et al. 2017) architectures. This will enable systematic investigation of how particular latent dimensions map onto colour, texture, and pattern features, providing insight into the structure of artificial camouflage spaces.

We have previously used artificial moth targets on natural tree bark as the study system; however we are open to explore alternative organisms or environments with well-characterised visual statistics. During this project, you will gain experience in deep learning, computer vision, and evolutionary modelling, while contributing to current debates on interpretability and representation in biologically inspired AI systems.

Three research papers in the area

1. Talas et al. (2020). CamoGAN: Evolving optimum camouflage with Generative Adversarial Networks <https://doi.org/10.1111/2041-210X.13334>

2. Cuthill (2019). Camouflage <https://doi.org/10.1111/jzo.12682>

3. <https://github.com/junyanz/pytorch-CycleGAN-and-pix2pix>

Biological systems involved, Area of study, Other key words associated with the project

Animal;Aquatic ecosystem;Terrestrial ecosystem; Evolution;Methods;Animal;
behaviour;Morphology;Exploratory; Protective colouration; Deep learning; Camouflage

Bioinformatics field, Biological data & Programming language(s) used

Machine Learning/AI;Comparative genomics; Images/video; Python;R; Custom written python scripts

Helpful previous knowledge

Previous coding experience (especially Python) and basic understanding of deep learning certainly comes handy, however is not a requirement. If you have no previous experience in coding, but have the aptitude to learn it while also interested in protective colouration, you are encouraged to apply!

Other relevant information

I'm based at the Vet School in Langford, which is a 40 minute commute from the main university campus with the U2 bus (ticket provided by the Vet School). There is no requirement to attend daily, but a weekly personal meeting is recommended and there will be some programming involved in local PCs.

21. Comparative Genomic Analysis of Symbiotic and Aposymbiotic States in Mollusks

Dr Vengamanaidu Modepalli

vengamanaidu.modepalli@bristol.ac.uk

General research field

Evolution

Project title

Comparative Genomic Analysis of Symbiotic and Aposymbiotic States in Mollusks

Project description

Symbiosis is a major driver of evolutionary innovation. In molluscan photosymbiosis, the animal host transitions from a purely predatory lifestyle to one supported by autotrophy. This transition requires significant genomic adaptation. This project aims to identify the genomic signatures of photosymbiosis by comparing the genomes of symbiotic and aposymbiotic (non-symbiotic or bleached) molluscs. While most molluscs are heterotrophic, certain lineages—most notably the giant clams (Tridacninae) and solar-powered sea slugs (Sacoglossa)—have evolved the ability to harbour photosynthetic partners or their organelles. Through comparative genomics, this study will investigate how the host genome has been "rewired" to facilitate a symbiotic relationship.

Three research papers in the area

1. Li, R., Leiva, C., Lemer, S. et al. Photosymbiosis shaped animal genome architecture and gene evolution as revealed in giant clams. *Commun Biol* 8, 7 (2025). <https://doi.org/10.1038/s42003-024-07423-8>
2. Ruiqi Li, Daniel Zarate, Viridiana Avila-Magaña, Jingchun Li; Comparative transcriptomics revealed parallel evolution and innovation of photosymbiosis molecular mechanisms in a marine bivalve. *Proc Biol Sci* 1 May 2024; 291 (2023): 20232408. <https://doi.org/10.1098/rspb.2023.2408>
3. Li R, Li J, Lopez JV et al. The genome sequence of the giant clam, *Tridacna gigas* (Linnaeus, 1758) [version 1; peer review: 2 approved, 1 approved with reservations]. *Wellcome Open Res* 2024, 9:145 (<https://doi.org/10.12688/wellcomeopenres.21136.1>)

Biological systems involved, Area of study, Other key words associated with the project

Eukaryote;Animal; Evolution;Genomic;Physiology

Bioinformatics field, Biological data & Programming language(s) used

Genomics;Comparative genomics;Metabolomics;Gene families; RNA;Proteins;DNA; R;Bash;OrthoVenn

22. Comparative Genomic Analysis of Symbiotic and Aposymbiotic States in Flatworms (Acoels)

Dr Vengamanaidu Modepalli

vengamanaidu.modepalli@bristol.ac.uk

General research field

Evolution

Project description

Acoel flatworms represent some of the most fundamental bilaterally symmetrical animals and are essential in exploring the origins of animal-algal symbiosis. Photosymbiotic acoels are born without their symbiotic partners and must actively seek out specific algae in the seawater for survival. Once these algae are internalized, the acoel's anatomy and metabolism undergo significant transformations: it ceases heterotrophic feeding and becomes entirely reliant on the products of algal photosynthesis. This project aims to uncover the permanent genomic adaptations that facilitate the evolution of photosymbiosis by comparing the genomes of obligate symbiotic acoels (such as *Symsagittifera roscoffensis*) with strictly heterotrophic, free-living acoels (like *Hofstenia miamia*). Through this research, we will delineate the essential "Photosymbiotic Toolkit" that differentiates solar-powered organisms from their predatory counterparts.

Three research papers in the area

1. Nanes Sarfati, D., Xue, Y., Song, E.S. et al. Coordinated wound responses in a regenerative animal-algal holobiont. *Nat Commun* 15, 4032 (2024). <https://doi.org/10.1038/s41467-024-48366-2>
2. Loss of immune signalling pathways and increased pathogen susceptibility associated to photosymbiosis in acoels. Francesca Pinton, Nadezhda N. Rimskaya Korsakova, Katja Felbel, Elisabeth Grimmer, Andreas Hejnol *bioRxiv* 2025.06.17.660074; doi: <https://doi.org/10.1101/2025.06.17.660074>
3. Pedro Martinez, Kirill Ustyantsev, Mikhail Biryukov, Stijn Mouton, Liza Glasenburg, Simon G Sprecher, Xavier Bailly, Eugene Berezikov, Genome assembly of the acoel flatworm *Symsagittifera roscoffensis*, a model for research on body plan evolution and photosymbiosis, *G3 Genes|Genomes|Genetics*, Volume 13, Issue 2, February 2023, jkac336, <https://doi.org/10.1093/g3journal/jkac336>

Biological systems involved, Area of study, Other key words associated with the project

Eukaryote;Animal; Evolution;Genomic;Physiology

Bioinformatics field, Biological data & Programming language(s) used

Comparative genomics;Gene families;Metabolomics; DNA;RNA;Proteins; R;Bash;OrthoFinder

23. Comparative Genomic Analysis of Symbiotic and Aposymbiotic Cnidarians

Dr Vengamanaidu Modepalli

vengamanaidu.modepalli@bristol.ac.uk

General research field

Evolution and comparative genomics

Project title

Comparative Genomic Analysis of Symbiotic and Aposymbiotic Cnidarians

Project description

Photosymbiosis represents an evolutionary "innovation" that has emerged independently across various branches of the animal kingdom. However, it remains unclear whether symbiotic animals simply repurpose existing genes or if they possess a distinct "symbiosis toolkit" that non-symbiotic animals do not have. Non-symbiotic species serve as an ideal "ancestral-like" baseline for understanding the genome of a standard, predatory anemone prior to the integration of algae. This project aims to explore the fundamental genomic shifts necessary for an organism to transition from a strictly predatory (heterotrophic) lifestyle to one that is solar-powered (autotrophic). While many cnidarians can temporarily expel their algae (a process known as bleaching), certain lineages have permanently adapted their genomes to accommodate photosynthetic partners. By comparing the genome of obligate symbiotic species with that of strictly non-symbiotic (aprosymbiotic) species, this study seeks to uncover the "evolutionary blueprint" of photosymbiosis.

Three research papers in the area

1. Cnidarian-Symbiodiniaceae symbiosis establishment is independent of photosynthesis Jinkerson, Robert E. et al. Current Biology, Volume 32, Issue 11, 2402 - 2415.e4
2. Cui, G., Mi, J., Moret, A. et al. A carbon-nitrogen negative feedback loop underlies the repeated evolution of cnidarian–Symbiodiniaceae symbioses. Nat Commun 14, 6949 (2023). <https://doi.org/10.1038/s41467-023-42582-y>
3. L.M. Gorman, T.R. Tivey, E.H. Raymond, I.A. Ashley, C.A. Oakley, A.R. Grossman, V.M. Weis, & S.K. Davy, Stability of the cnidarian–dinoflagellate symbiosis is primarily determined by symbiont cell-cycle arrest, Proc. Natl. Acad. Sci. U.S.A. 122 (14) e2412396122, <https://doi.org/10.1073/pnas.2412396122> (2025).

Biological systems involved, Area of study, Other key words associated with the project

Animal; Genomic; Evolution; Physiology

Bioinformatics field, Biological data & Programming language(s) used

Genomics; Comparative genomics; Phylogenetics/phylogenomics; RNA; DNA; Proteins; R; Bash; OrthoVenn

24. The Geographical Distribution and Ecology of *Synechococcus* sp. PCC 7336, potentially one of the first cyanobacteria to invade the surface ocean

Dr Joanne Boden and Prof Patricia Sánchez-Baracaldo

j.boden@bristol.ac.uk

General research field

Evolutionary microbiology / co-evolution of Earth and life on geological time scales

Project description

In this project, we will conduct phylogenetic analyses and biogeographical analyses on a specific strain of cyanobacteria called *Synechococcus* sp. PCC 7336. *Synechococcus* sp. PCC 7336 is an ancient strain that emerged around 2.1 billion years ago when the planet was still relatively young – for context the planet formed from a cloud of space dust 4.6 billion years ago, and the first life emerged somewhere between 3.55 and 4.40 billion years ago. Animals and plants emerged within the last 0.9 billion years. Cyanobacteria are found in a wide variety of habitats in the world today, ranging from oceans to deserts, lakes, ice caps, and soil, but *Synechococcus* sp. PCC 7336 is exciting because it may represent one of only four clades which can be found in the open ocean. Unlike other marine cyanobacteria, we know relatively little about PCC 7336 because it sits on its own on a long branch in the cyanobacterial tree of life. Yet because of this long branch, it offers the potential to inform us on how cyanobacteria invaded the ocean hundreds of millions of years ago. The aims of this project are three-fold: 1) Find close relatives of PCC 7336 which haven't had their genomes sequenced by performing BlastN analyses on publicly available databases and reconstructing a phylogenetic tree from 16S rRNA. 2) Establish which habitats the closest relatives of PCC 7336 have been found in to determine whether PCC 7336 is part of a larger marine clade or not 3) Estimate whether PCC 7336 is restricted to coastal environments or more widespread in the open ocean by generating a map of the global geographical distribution of PCC 7336. This will be achieved by performing sequence searches in publicly available microbial gene catalogues to establish how widespread it is in marine environments today, as a potential proxy of where its ancestors might have lived in the past.

Three research papers in the area

SANCHEZ-BARACALDO, P., BIANCHINI, G., WILSON, J. D. & KNOLL, A. H. 2022. Cyanobacteria and biogeochemical cycles through Earth history. *Trends in Microbiology*, 30, 143-157.

SANCHEZ-BARACALDO, P. 2015. Origin of marine planktonic Cyanobacteria. *Scientific Reports*, 5, 17418.

BIANCHINI, G., HAGEMANN, M. & SÁNCHEZ-BARACALDO, P. 2024. Stochastic Character Mapping, Bayesian Model Selection, and Biosynthetic Pathways Shed New Light on the Evolution of Habitat Preference in Cyanobacteria. *Systematic Biology*, 73, 644-665.

Biological systems involved, Area of study, Other key words associated with the project

Prokaryote; Aquatic ecosystem; Exploratory; Evolution; Genomic; Ecology; Geobiology, Astrobiology

Bioinformatics field, Biological data & Programming language(s) used

Genomics;Metagenomics;Phylogenetics/phylogenomics;Database searching;NGS data; DNA; Bash; IQ-TREE, MAFFT, TrimAl, BlastN

Helpful previous knowledge

Cyanobacteria were one of the first organisms to release oxygen into Earth's atmosphere. They do this by performing oxygenic photosynthesis. All plants and algae owe their photosynthetic capabilities to an ancestral cyanobacterium which was engulfed by an ancestral eukaryote and later gave rise to chloroplasts. Therefore, it is thanks to cyanobacteria and their descendants, that our planet developed an atmosphere that was breathable for complex eukaryotes including animals, plants, and fungi. Previous research has estimated that other groups of marine planktonic cyanobacteria evolved around 1.0 to 0.6 billion years ago, shortly before a noteworthy rise in atmospheric oxygen concentrations, known as the Neoproterozoic Oxygenation Event.

25. Dating the Evolution of Nutrient Acquisition Strategies in Ancient Bacteria

Dr Joanne Boden and Prof Patricia Sánchez-Baracaldo

j.boden@bristol.ac.uk

General research field

Evolutionary microbiology / Co-evolution of Earth and life on geological timescales

Project description

Phosphorus is one of the six key elements required for life (the others being carbon, hydrogen, nitrogen, oxygen and sulfur – small amounts of transition metals are also required), so there has been much research in recent years aimed at understanding whether planetary supplies of phosphorus limited the early spread and diversification of life between 3.2 and 2.5 billion years ago – for context Earth formed 4.6 billion years ago and primitive animals emerged within the last 0.9 billion years. Phylogenomic analyses recently estimated that environmental phosphate concentrations declined during this time (Boden et al., 2024), potentially affecting the growth and development of early micro-organisms. Some of these early micro-organisms were the ancestors of modern-day Cyanobacteria. Cyanobacteria are the only prokaryotes capable of oxygenic photosynthesis. Oxygenic photosynthesis is the only biological metabolism that can release substantial volumes of oxygen into the environment, so cyanobacteria and their descendants were likely responsible for oxygenating the planet's atmosphere, priming it for the development of complex eukaryotic life. Given their important role in oxygenating the atmosphere, it would be useful to know how and when Cyanobacteria acquired mechanisms (i.e. genes) to access phosphorus for growth and development.

Phosphorus can be acquired in the form of phosphate (an inorganic molecule often measured to get a sense of how much phosphorus there is in modern day habitats), phosphonate, or phosphate esters. The addition of genes allowing cyanobacteria to access each different source would have widened cyanobacteria's arsenal of phosphorus-uptake strategies, potentially giving them the ability to grow and develop on different molecules and in different environments. In this project, you will estimate when cyanobacteria acquired genes to access phosphorus from phosphate, phosphonate, and phosphate esters by conducting sequence searches (ideally with HMMer), reconstructing phylogenetic trees of each gene, and then reconciling them with a time-calibrated species tree.

Three research papers in the area

SANCHEZ-BARACALDO, P., BIANCHINI, G., WILSON, J. D. & KNOLL, A. H. 2022. Cyanobacteria and biogeochemical cycles through Earth history. *Trends in Microbiology*, 30, 143-157.

BODEN, J. S., ZHONG, J., ANDERSON, R. E. & STÜEKEN, E. E. 2024. Timing the evolution of phosphorus-cycling enzymes through geological time using phylogenomics. *Nature Communications*, 15, 3703.

WILLIAMS, T. A., DAVIN, A. A., SZÁNTHÓ, L. L., STAMATAKIS, A., WAHL, N. A., WOODCROFT, B., SOO, R. M., EME, L., SHERIDAN, P. O., GUBRY-RANGIN, C., SPANG, A., HUGENHOLTZ, P. & SZÖLLOSI, G. J. 2024. Phylogenetic reconciliation: making the most of genomes to understand microbial ecology and evolution. *Isme Journal*, 18, 14.

Biological systems involved, Area of study, Other key words associated with the project

Prokaryote; Exploratory; Evolution; Genomic; Ecology; Geobiology, Astrobiology

Bioinformatics field, Biological data & Programming language(s) used

Genomics; Comparative genomics; Phylogenetics/phylogenomics; DNA; Python; Bash; AleRax, IQ-TREE, MAFFT, TrimAl, HMMer

Helpful previous knowledge

Cyanobacteria were one of the first organisms to release oxygen into Earth's atmosphere. They do this by performing oxygenic photosynthesis. All plants and algae owe their photosynthetic capabilities to an ancestral cyanobacterium which was engulfed by an ancestral eukaryote and later gave rise to chloroplasts. Therefore, it is thanks to cyanobacteria and their descendants, that our planet developed an atmosphere that was breathable for complex eukaryotes including animals, plants, and fungi. Previous research has estimated that other groups of marine planktonic cyanobacteria evolved around 1.0 to 0.6 billion years ago, shortly before a noteworthy rise in atmospheric oxygen concentrations, known as the Neoproterozoic Oxygenation Event.

26. Integrative Omics Analysis of the Tumour Microenvironment

Dr Daniel D'Andrea

daniel.dandrea@bristol.ac.uk

General research field

Computational Cancer Biology

Project description

The tumour microenvironment (TME) is a dynamic, complex ecosystem where cancer cells interact with non-cancerous cells and stromal components, significantly influencing tumour progression and therapy response. Understanding the mechanisms underlying this complexity is crucial for identifying potential therapeutic targets.

This project aims to harness data-driven analysis on diverse multi-omics datasets of solid cancers (such as non-small cell lung cancer, colon-rectal cancer, ovarian cancer) alongside clinical patient data, with the goal of revealing cellular dynamics within the TME and identifying key genes influencing cancer progression and therapy response.

The project will involve developing and evaluating deep learning and/or machine learning models, combined with bioinformatics tools, to 1) create high-resolution maps of cellular subpopulations and spatial structures; 2) analyse these maps of cell subpopulations to identify factors and sources of perturbation impacting tumour progression and therapeutic response.

The student will work within an interdisciplinary research environment, collaborating with immunologists, bioinformaticians, and data scientists. Additional project ideas in the area of omics and cancer are available within the group and can be discussed.

Three research papers in the area

1- Sidiropoulos D N, et al, Neoadjuvant Immunotherapy Promotes the Formation of Mature Tertiary Lymphoid Structures in a Remodeled Pancreatic Tumor Microenvironment. Cancer Immunol Res (2025) 13 (11): 1716–1731. <https://doi.org/10.1158/2326-6066.CIR-25-0387>

2- Singhal, V., Chou, N., Lee, J. et al. BANKSY unifies cell typing and tissue domain segmentation for scalable spatial omics data analysis. Nat Genet 56, 431–441 (2024). <https://doi.org/10.1038/s41588-024-01664-3>

3- Jin, S., Plikus, M.V. & Nie, Q. CellChat for systematic analysis of cell–cell communication from single-cell transcriptomics. Nat Protoc 20, 180–219 (2025). <https://doi.org/10.1038/s41596-024-01045-4>

Biological systems involved, Area of study, Other key words associated with the project

Human; Medical/clinical;Methods

Bioinformatics field, Biological data & Programming language(s) used

Transcriptomics;Machine Learning/AI;NGS data; RNA; R

Helpful previous knowledge

R programming

27. Quantifying the Aunts and Uncles Effect in parentage analysis

Dr Hugo Harrison

hugo.harrison@bristol.ac.uk

Other research staff, PhD students of technicians involved (if applicable)

Rebecca Loy

General research field

Population genetics

Project description

Parentage analysis is a core component of population genetics, conservation biology, and breeding program management. Traditional parentage assignment focuses on identifying true parents from genotypic data; however, in complex populations close relatives can complicate or bias parentage inference and pedigree reconstructions. This phenomenon, referred to as the aunts and uncles effect, arises because siblings share a high proportion of alleles with the true parents, increasing the likelihood of false assignments or ambiguous parentage calls when genotypic resolution is limited. This project aims to quantify the aunts and uncles effect in parentage analysis and pedigree reconstruction. Using simulated datasets and empirical genotype data, the project will examine how the presence of close relatives in the candidate pool influences assignment accuracy, likelihood scores, and confidence thresholds.

The literature review will focus on the uses of parentage analysis and pedigree reconstruction in conservation genetics. The project component will evaluate the performance of Sequoia, a powerful parentage analysis tool, under controlled scenarios where aunts and uncles are deliberately included in the candidate set.

This project is suited to candidates with a strong desire to develop applied coding skills in R, an interest in population genetics, and can independently learn to use R based software packages (e.g., Sequoia).

Three research papers in the area

Olsen, J. B., Busack, C., Britt, J., & Bentzen, P. (2001). The Aunt and Uncle Effect: An Empirical Evaluation of the Confounding Influence of Full Sibs of Parents on Pedigree Reconstruction. In *The American Genetic Association* (Vol. 92).

Harrison, H. B., Saenz-Agudelo, P., Planes, S., Jones, G. P., & Berumen, M. L. (2013). Relative accuracy of three common methods of parentage analysis in natural populations. *Molecular Ecology*, 22(4). <https://doi.org/10.1111/mec.12138>

Huisman, J. (2017). Pedigree reconstruction from SNP data: parentage assignment, sibship clustering and beyond. *Molecular Ecology Resources*, 17(5), 1009–1024. <https://doi.org/10.1111/1755-0998.12665>

Biological systems involved, Area of study, Other key words associated with the project
quantitative simulations; Methods

Bioinformatics field, Biological data & Programming language(s) used
Population genomics;Barcoding; DNA; R; R packages: Sequoia

28. Establishing the provenance of museum specimens of Kākā and Kākāpō

Dr Hugo Harrison

hugo.harrison@bristol.ac.uk

General research field

population genetics, conservation

Project description

The origins of natural history collections are intertwined with colonial and extractive narratives. 'Specimens'; animal bodies, were often collected without consent or consideration for the deep cultural significance they held for Indigenous communities. In this project you will explore the provenance of specimens of kākāpō and kākā currently held in the World Wildlife Gallery at Bristol Museum & Art Gallery (BMAG). Tissues of museums specimens will be extracted, genotyped using known microsatellite markers, and compared to specimens of known origins to establish the likely provenance of specimens held at BMAG.

The literature review will focus on the use of population genetics to establish provenance in conservation research. The project component will develop pipelines for genetic assignment tests and geographic assignments using microsatellite data.

This project is suited to candidates with a desire to use bioinformatics skills in conservation, an interest in population genetics, and can independently learn to use R based software packages (e.g., adegenet, scatR).

Three research papers in the area

Bergner, L. M., Dussex, N., Jamieson, I. G., & Robertson, B. C. (2016). European colonization, not polynesian arrival, impacted population size & genetic diversity in the critically endangered New Zealand Kakapo. *Journal of Heredity*, 107(7), 593–602. <https://doi.org/10.1093/jhered/esw065>

Sainsbury, J. P., Greene, T. C., Moorhouse, R. J., Daugherty, C. H., & Chambers, G. K. (2006). Microsatellite analysis reveals substantial levels of genetic variation but low levels of genetic divergence among isolated populations of Kaka (*Nestor meridionalis*). *Emu - Austral Ornithology*, 106(4), 329–338. <https://doi.org/10.1071/MU06009>

Wasser, S. K., Shedlock, A. M., Comstock, K., Ostrander, E. A., Mutayoba, B., & Stephens, M. (2004). Assigning African elephant DNA to geographic region of origin: Applications to the ivory trade. www.pnas.org/cgi/doi/10.1073/pnas.0403170101

Biological systems involved, Area of study, Other key words associated with the project

Animal; Methods; Ecology

Bioinformatics field, Biological data & Programming language(s) used

Population genomics; DNA; R; R packages: adegenet, scatR

29. Genome-wide mapping of genetic variation in a wild anemonefish population

Dr Hugo Harrison

hugo.harrison@bristol.ac.uk

Other research staff, PhD students of technicians involved (if applicable)

Rebecca Loy

General research field

Comparative genomics

Project description

Genome mapping is a foundational tool in genetics, evolution, and biomedical research that allows us to connect genomic variation to biological function, evolutionary processes, and phenotypic traits. This project aims to map patterns of genetic variation across the genome of a wild anemonefish population using SNP data and an available reference genome for *Amphiprion percula*. By integrating variant calling, diversity analysis, and structural variant detection, the study will produce a high-resolution genomic landscape highlighting how selection, recombination, and demographic history shape genetic variation in a wild population of *A. percula* in Kimbe Bay.

The literature review could focus on the applications of genomic tools to detect inbreeding and manage small populations or the role of genomic diversity on genome architecture. The project will use genome-wide SNP data from 96 individuals and a high-quality reference genome to (i) map population-level variation across chromosomes, (ii) detect genomic regions of unusually high/low diversity, (iii) infer linkage and recombination patterns, and (iv) identify candidate structural variants or chromosomal inversions that shape genomic architecture.

This project is suited to candidates with an interest in genetics or evolutionary biology, and a desire to develop genomic analysis pipelines independently (e.g., SAMtools, Bowtie2).

Three research papers in the area

Almany, G. R., Planes, S., Thorrold, S. R., Berumen, M. L., Bode, M., Saenz-Agudelo, P., Bonin, M. C., Frisch, A. J., Harrison, H. B., Messmer, V., Nanninga, G. B., Priest, M. A., Srinivasan, M., Sinclair-Taylor, T., Williamson, D. H., & Jones, G. P. (2017). Larval fish dispersal in a coral-reef seascape. *Nature Ecology and Evolution*, 1(6). <https://doi.org/10.1038/s41559-017-0148>

Lehmann, R., Lightfoot, D. J., Schunter, C., Michell, C. T., Ohyanagi, H., Mineta, K., Foret, S., Berumen, M. L., Miller, D. J., Aranda, M., Gojobori, T., Munday, P. L., & Ravasi, T. (2019). Finding Nemo's Genes: A chromosome-scale reference assembly of the genome of the orange clownfish *Amphiprion percula*. *Molecular ecology resources*, 19(3), 570–585. <https://doi.org/10.1111/1755-0998.12939>

Biological systems involved, Area of study, Other key words associated with the project

Animal; Genomic

Bioinformatics field, Biological data & Programming language(s) used

_Genomics; Population genomics; Comparative genomics; DNA; R; Bash

Differentially expressed genes in circadian centres of the mammalian brain.

30. Differentially expressed genes in circadian centres of the mammalian brain

Prof Hugh Piggins and Dr Lukasz Chrobok

hugh.piggins@bristol.ac.uk

General research field

Circadian rhythms

Project title

Differentially expressed genes in circadian centres of the mammalian brain.

Project description

The body's main circadian centre is contained within the suprachiasmatic nuclei (SCN) of the brain's hypothalamus. The molecular basis of the SCN clock has been determined with a transcriptional-translational feedback/forward loop (TTFL) model appearing to be evolutionarily conserved. Key genes/proteins in this SCN TTFL are Per1-2/PER1-PER2, Arntl/Bmal1, and Nr1d2, NR1D2. Intriguingly, these genes/proteins are also expressed in other parts of the brain, suggesting that there are extra-SCN TTFL oscillators. In this project, the student will use bioinformatic approaches to evaluate daily rhythms in the TTFL as well as other non-clock genes in two brain centres, the medial habenula (MHb) and the dorsal vagal complex (DVC), both of which have been hypothesized to be candidate extra-SCN TTFL oscillators. The project will use existing RNAseq datasets to examine similarities and differences between the MHb and DVC. The student will also evaluate a number of online R packages to ascertain which package is the best suited for evaluating rhythms in gene expression in extra-SCN oscillators.

Three research papers in the area

_Hitrec et al., iScience 2023 26(2):106002.

Liu et al., Trends Neurosci. 2025 Nov;48(11):904-918.

Curtis and Piggins Eur. J. Neurosci. 2024 60(2):3921-3945.

Biological systems involved, Area of study, Other key words associated with the project

_Eukaryote; Model organism; Physiology; Neuroscience; Genomic; mouse; sleep; daily rhythms

Bioinformatics field, Biological data & Programming language(s) used

Transcriptomics; RNA; DNA; R; diffCircadian, DeqSeq2.

Helpful previous knowledge

Some knowledge of brain anatomy, circadian rhythms and sleep.

31. AI-powered Design of ADDomer Nanoparticle-based Synthetic Supervaccines

Prof Imre Berger and Prof Christiane Schaffitzel

imre.berger@bristol.ac.uk

Other research staff, PhD students of technicians involved (if applicable)

Dr Georgia Balchin, Milo Friedrich, Jonathan Light

General research field

Vaccinology, AI, protein engineering, rational design, synthetic biology, computational biology, molecular interactions

Project title

AI-powered Design of ADDomer Nanoparticle-based Synthetic Supervaccines

Project description

Protein nanoparticles offer powerful platforms for multivalent antigen display, but the precise and stable attachment of complex, trimeric viral glycoproteins remains a major design challenge. In our laboratory, we have developed a novel virus-inspired system based on a nanoparticle we discovered, the ADDomer (adenovirus-derived dodecahedron). The ADDomer consists of 60 copies of an engineered adenoviral penton base protein (PBP), arranged into 12 pentons that self-assemble into a megadalton-sized dodecahedral nanoparticle. ADDomer is highly thermostable and excellently suited for high-density antigen display, a property we have previously exploited for vaccine design [1].

In native adenovirus, the pentons serve as attachment sites for the viral fibers. Using AI-based protein structure prediction and engineering, we have re-designed this fiber–penton interaction into a modular molecular adaptor. This adaptor allows the multivalent display of entire proteins, including antibodies, on the ADDomer surface, resulting in ultra-high-avidity constructs termed “Gigabodies” [2].

In this project, we now aim to further advance this platform using a sophisticated computational design strategy. The student will take the next step by rationally converting a high-affinity but non-covalent protein–protein interface into a site-specific covalent interaction. Using structural modelling and AI-driven protein design tools (e.g. AlphaFold-based modelling [3], interface analysis, and cysteine scanning), the student will identify complementary residue pairs on the adenoviral penton cleft and the engineered adaptor that can be mutated to cysteines to enable stabilising disulfide bond formation upon assembly.

The project combines structural bioinformatics, protein interface design, and virology-inspired nanotechnology, with direct experimental validation in our laboratory. Successful candidates will see their in silico designs tested first using fluorescent reporter proteins and subsequently with medically relevant viral glycoproteins, including influenza haemagglutinin (HA), SARS-CoV-2 spike, and rabies virus glycoprotein. The long-term goal is to generate improved, highly stable nanoparticle-based “supervaccines”.

This project is ideal for students interested in AI-driven protein engineering, structural biology, and translational vaccine or biologics design.

Three research papers in the area

- [1] Vragliau C et al. Synthetic self-assembling ADDomer platform for highly efficient vaccination by genetically encoded multiepitope display. *Science Advances* 2019 Sep 25;5(9):eaaw2853. doi: 10.1126/sciadv.aaw2853.
- [2] Buzas D et al. In vitro generated antibodies guide thermostable ADDomer nanoparticle design for nasal vaccination and passive immunization against SARS-CoV-2. *Antibody Therapy*, 2023 Oct 17;6(4):277-297. doi: 10.1093/abt/tbad024.
- [3] Jumper J et al. Highly accurate protein structure prediction with AlphaFold. *Nature*. 2021 Aug;596(7873):583-589. doi: 10.1038/s41586-021-03819-2. Epub 2021 Jul 15. PMID: 34265844

Biological systems involved, Area of study, Other key words associated with the project

Eukaryote;Human;Viral glycoproteins (SARS-CoV-2 Spike, flu HA, Rabies virus glycoprotein),

Exploratory;Medical/clinical;Treatment/Therapeutics;Development;Biochemistry;Synthetic Biology;Structural Biology;Vaccinology; Structural bioinformatics; AI-driven protein design; AlphaFold-based modelling; Protein–protein interface analysis; Computational mutagenesis; Disulfide bond engineering; Nature-inspired nanotechnology

Bioinformatics field, Biological data & Programming language(s) used

Machine Learning/AI;Bioengineering;Protein design; Proteins;protein interactions; Python AlphaFold, RF Diffusion, Pymol, Chimera, other recent structure prediction and protein design software

Helpful previous knowledge

General knowledge about protein structure; basic biological chemistry (amino acids, disulfide bonds); knowledge about protein structure prediction and design software (AlphaFold, RFdiffusion etc).

Other relevant information

Our lab is a multidisciplinary synthetic and structural biology group at the School of Biochemistry, part of the Max Planck Bristol Centre. We combine protein engineering, bioinformatics, structural biology, and nanotechnology to tackle cutting-edge problems in fundamental biology, protein therapeutics and vaccine development.

We maintain very close collaborations with Christiane Schafftzell's Cryo-EM lab, providing students with exposure to state-of-the-art structural techniques and integrative approaches. Our team is diverse and inclusive, fostering a supportive environment where students are encouraged to learn, innovate, and contribute meaningfully to ongoing projects.

We have a “work hard, play hard” attitude — celebrating successes, learning from challenges, and enjoying team events. Students will gain hands-on experience in computational protein design, structural modelling, and translational research, building skills directly applicable to academic and industrial careers.

32. Spatial Transcriptomic Analysis of Plant Immune Regulator PDE1 (Using Single-Cell RNA-seq)

Dr Pierre Buscaill and Maria Sachkova

pierre.buscaill@bristol.ac.uk

General research field

Plant Molecular Biology / Bioinformatics

Project description

This project aims to characterize the spatial expression patterns of the plant immune regulator PDE1 using single-cell (sc)RNA sequencing data. Students will analyse publicly available scRNA-seq datasets to identify cell-type-specific expression, spatial heterogeneity, and co-expression networks of PDE1. This work will provide insights into PDE1's role in plant immunity and its regulation across different tissue types.

Three research papers in the area

Nobori et al. (2025) A rare PRIMER cell state in plant immunity. *Nature*, 638, 197–205.
<https://doi.org/10.1038/s41586-024-08383-z>

Tang, Bozeng et al. (2023) Cell-type-specific responses to fungal infection in plants revealed by single-cell transcriptomics. *Cell Host & Microbe*, Volume 31, Issue 10, 1732 - 1747.e5
<https://doi.org/10.1016/j.chom.2023.08.019>

Isner et al. (2019) Short- and Long-Term Effects of UVA on Arabidopsis Are Mediated by a Novel cGMP Phosphodiesterase. *Curr Biol.* (15):2580-2585.e4. <https://doi.org/10.1016/j.cub.2019.06.071>

Biological systems involved, Area of study, Other key words associated with the project

Eukaryote;Plants;Agriculture;Model organism;Microbes (bacteria);Prokaryote;
Exploratory;Agriculture;Plant immunity;Development;Genomic;Physiology; Single-cell RNA-seq, plant immunity, spatial transcriptomics, PDE1

Bioinformatics field, Biological data & Programming language(s) used

Transcriptomics;NGS data;Data Modeling; RNA;cDNA; Python;R; RStudio / Python

Helpful previous knowledge

R or Python programming, familiarity with RNA-seq data, Basic molecular biology

Other relevant information

Opportunity to combine computational analysis with biological interpretation; suitable for students interested in both bioinformatics and plant biology

33. AI-Driven Discovery of Bacterial Strategies to Manipulate Host Immune Signalling

Dr Pierre Buscaill and Maria Sachkova

pierre.buscaill@bristol.ac.uk

General research field

Plant-Microbe Interactions / Bioinformatics

Project title

AI-Driven Discovery of Bacterial Strategies to Manipulate Host Immune Signalling

Project description

This project uses AI-based approaches to investigate how bacterial pathogens may manipulate host immune signalling across kingdoms. The focus is on cyclic GMP (cGMP), an important signalling molecule in plant immunity, and on PDE1, a plant protein that regulates cGMP levels.

The student will primarily work with protein sequence analysis, structure prediction, and AI-based modelling tools to explore whether bacterial effectors interfere with this pathway. The project is exploratory and hypothesis-driven, and does not require prior expertise in plant immunity.

The project has three main computational objectives:

1. Predict and analyse protein structures of bacterial effectors and the plant protein PDE1 using AlphaFold, followed by structural comparison and interaction analysis using AlphaFold and ChimeraX.
2. Search bacterial secreted protein datasets for phosphodiesterase-like domains using sequence similarity, domain databases, and profile-based methods, to identify effectors that may mimic plant PDE1 protein.
3. Integrate sequence and structural information to predict potential effector functions and generate testable hypotheses about how pathogens manipulate host immune signalling.

This project is well suited for students interested in AI-driven biology, protein structure prediction, and bioinformatics discovery. The emphasis is on learning and applying computational tools to real biological problems.

Three research papers in the area

1. Kwiatkowski et al. (2024) Cyclic nucleotides – the rise of a family. Trends in Plant Science. Volume 29, Issue 8, August 2024, Pages 915-924. <https://doi.org/10.1016/j.tplants.2024.02.003>
2. Yu et al. (2022) TIR domains of plant immune receptors are 2',3'-cAMP/cGMP synthetases mediating cell death. Volume 185, Issue 13, 23 June 2022, Pages 2370-2386.e18. <https://doi.org/10.1016/j.cell.2022.04.032>
3. Isner et al. (2019) Short- and Long-Term Effects of UVA on Arabidopsis Are Mediated by a Novel cGMP Phosphodiesterase. Current Biology. (15):2580-2585.e4. <https://doi.org/10.1016/j.cub.2019.06.071>

Biological systems involved, Area of study, Other key words associated with the project

Prokaryote;Eukaryote;Plants;Agriculture;Model organism;Bacteria; Structural; Biology;Biochemistry;Plant immunity; structural bioinformatics, bacterial effectors, cGMP signalling, PDE1, plant-pathogen interactions

Bioinformatics field, Biological data & Programming language(s) used

Proteomics;Machine Learning/AI;Data Modeling;Database searching; Proteins; Python;R;Bash; AlphaFold, ChimeraX, BLAST, HMMER, Python packages for structural analysis (Biopython)

Helpful previous knowledge

R/Python programming, familiarity with BLAST, sequence alignment, structural visualization, molecular biology, protein structure

Other relevant information

This project is entirely computational and provides hands-on experience in AI-based protein structure prediction, structural bioinformatics, and sequence analysis applied to a real biological question. It is ideal for students interested in computational biology, protein modelling, and hypothesis-driven bioinformatics, regardless of prior background in plant immunity.

34. Modelling and investigating evidence of natural selection in hybridising populations

Prof Mark Beaumont

mamab@bristol.ac.uk

General research field
Population Genomics

Project description

Hybridising populations provide a useful testbed for identifying the effects of natural selection. Genetic variants that are advantageous in one population may be disadvantageous in another and vice versa. Genomic methods can be used to measure the degree to which an individual has hybrid ancestry. In admixing populations we expect the chances of admixed cats having a neutral variant (equally fit in either population) to be in direct proportion to the admixture coefficient of the cat, given the difference in variant frequency between populations. Deviations from this expectation may indicate the effect of natural selection. There are a number of such genome-cline methods available, including one currently in development by my group.

Depending on student interest we could develop simulation tests to evaluate the performance of genome-cline methods using tools such as msprime and ADMIXTURE (using Python-based AdmixPipe). Alternatively we can use genome-cline methods to identify genes that are candidates for selection using real data. For the latter objective we have data on Scottish wildcats that have varying degrees of admixture with domestic cats. We can then apply functional analysis methods based on gene ontology and knowledge of gene regulatory networks to see what traits are being selected for.

Three research papers in the area

Howard-McCombe, J., Jamieson, A., Carmagnini, A., Russo, I.R.M., Ghazali, M., Campbell, R., Driscoll, C., Murphy, W.J., Nowak, C., O'Connor, T. and Tomsett, L., 2023. Genetic swamping of the critically endangered Scottish wildcat was recent and accelerated by disease. *Current Biology*, 33(21), pp.4761-4769.

Baumdicker, F., Bisschop, G., Goldstein, D., Gower, G., Ragsdale, A.P., Tsambos, G., Zhu, S., Eldon, B., Ellerman, E.C., Galloway, J.G. and Gladstein, A.L., 2022. Efficient ancestry and mutation simulation with msprime 1.0. *Genetics*, 220(3), p.iyab229.

Mussmann, S.M., Douglas, M.R., Chafin, T.K. and Douglas, M.E., 2020. AdmixPipe: population analyses in Admixture for non-model organisms. *Bmc Bioinformatics*, 21(1), p.337.

Biological systems involved, Area of study, Other key words associated with the project
Eukaryote;Terrestrial ecosystem;Animal; Exploratory;Evolution;Genomic;Methods

Bioinformatics field, Biological data & Programming language(s) used
Genomics;Population genomics;Data Modeling; DNA; Python;R;Bash; msprime; ADMIXTURE; AdmixPipe; vcftools

Helpful previous knowledge

Ideally some background in population genetics would be useful.

35. Predicting brain intercellular communications pathways from single cell RNAseq data

Prof David Murphy and Dr Michael Greenwood

d.murphy@bristol.ac.uk

Other research staff, PhD students of technicians involved (if applicable)

Benjamin Gillard

General research field

Biological predictions inferred from single cell transcriptome data

Project description

The supraoptic nucleus (SON) of the hypothalamus contains specialised peptidergic magnocellular neurones (MCNs) that project axons to the posterior pituitary gland neuro-vascular interface. To counteract hyperosmotic imbalance, such as that evoked by water deprivation or salt intake, MCNs respond to intrinsic osmosensory mechanisms and to excitatory inputs from osmosensitive neurones in the subfornical organ (SFO) by secreting neuropeptides such as arginine vasopressin (AVP) and oxytocin (OXT) into the bloodstream. AVP and OXT interact with specific kidney collecting duct receptors to promote water conservation and salt excretion respectively.

The distinct anatomical separation of neuropeptide synthesis in hypothalamic cell bodies and secretion from distal terminals in the posterior pituitary advantaged this system for studies which established our knowledge of the fundamental neuronal processes of stimulus-synthesis-secretion coupling. However, these advances mislead us into considering SON neurones as being simple factories churning out vast quantities of neuropeptides into the blood. Recent progress has revolutionised our view of the SON and illuminates its complex integrative functions.

Firstly, we have learnt that MCNs not only signal to the periphery via the posterior pituitary but are also engaged in intracerebral intercellular signalling that ensure coordination of hormonal and behavioural responses to challenges. MCNs send collateral axonal projections to numerous central nervous system targets that regulate perception, affective/emotional states and behaviours, including the hippocampus.

Secondly, our single nucleus RNA sequence analysis has revealed a remarkable diversity of SON cell types, with 10 being evident. Even more remarkable is the diversity of MCN cell-types revealed by single nucleus RNA sequence analysis. We confirmed that AVP and OXT expression is confined to distinct sub-populations of MCNs, but most unexpected was our revelation of a total of 8 different MCN subtypes in the SON.

This project will explore phenotypic diversity of the SON in the context of its different intercellular signalling modalities. The student will use an algorithm called CellChat to identify ligand-receptor pairs in single cell RNAseq data in order to predict intercellular signalling modalities within the SON, and also between the SON and other brain regions, such as the hippocampus.

Three research papers in the area

Jin S, Guerrero-Juarez CF, Zhang L, Chang I, Ramos R, Kuan CH, Myung P, Plikus MV, Nie Q. Inference and analysis of cell-cell communication using CellChat. Nat Commun. 2021 Feb 17;12(1):1088. doi:

10.1038/s41467-021-21246-9. PMID: 33597522; PMCID: PMC7889871.

Jin S, Plikus MV, Nie Q. CellChat for systematic analysis of cell-cell communication from single-cell transcriptomics. Nat Protoc. 2025 Jan;20(1):180-219. doi: 10.1038/s41596-024-01045-4. Epub 2024 Sep 16. PMID: 39289562.

Pauža AG, Mecawi AS, Paterson A, Hindmarch CCT, Greenwood M, Murphy D, Greenwood MP. Osmoregulation of the transcriptome of the hypothalamic supraoptic nucleus: A resource for the community. J Neuroendocrinol. 2021 Aug;33(8):e13007. doi: 10.1111/jne.13007. Epub 2021 Jun 22. PMID: 34297454.

Biological systems involved, Area of study, Other key words associated with the project
Animal; Model organism; Neuroscience; Physiology

Bioinformatics field, Biological data & Programming language(s) used
Transcriptomics; RNA; R; Python; Bash; CellChat - see references above

36. Exploring the role of AI-based structural prediction to improve AMR resistance prediction from bacterial genomes

Dr Sion Bayliss and Dr Jon Lees

s.bayliss@bristol.ac.uk

Other research staff, PhD students of technicians involved (if applicable)

Caroline Schreiber

General research field

Genomic AI/ML

Project description

This project focuses on the potential utility provided by protein structural embeddings as a predictor of antimicrobial resistance (AMR) in bacterial genomes. The core aim is to determine whether incorporating 3D structural information from protein complexes can improve AMR prediction compared to using nucleotide or amino acid sequences alone. The project will use *Mycobacterium tuberculosis* (Tb) as an exemplar pathogen.

Given uncertainties about AlphaFold's ability to accurately capture the structural effects of single nucleotide polymorphisms (SNPs), the project will begin with a comprehensive review of mutant effect types and alternative approaches which do not require training, such as zero-shot learning methods, before transitioning to the use of AlphaFold for prediction of structural variants a range of gene involve in quinolone resistance in Tb. These gene will include gyrA and gyrB which encode subunits of DNA gyrase, a type II topoisomerase that forms a functional heterotetramer (A₂B₂) with DNA and is the primary target for many fluoroquinolone antibiotics (including ciprofloxacin, levofloxacin, and moxifloxacin). Our hypothesis is that modelling this complete quaternary structure using AlphaFold 3 and extracting structural features may provide additional predictive power for AMR classification beyond existing sequence-based embeddings such as including ESM protein language model embeddings.

Initially, the student will conduct a literature review of mutant effect types in AMR-associated proteins, focusing on mechanisms by which SNPs confer resistance, and evaluate zero-shot and low-training approaches that could serve as baselines or alternatives to structure-based methods. Following this foundational work, the student will perform small-scale validation to determine whether AlphaFold 3 produces structurally sensible predictions when modelling known resistance mutations in the gyrA/gyrB complex, then extracting and engineering structural features from these predicted complexes. The student will compare predictive performance across different embedding types—nucleotide-based, amino acid-based, ESM embeddings, and novel structural embeddings derived from AF3 models—and develop machine learning classifiers for fluoroquinolone resistance prediction to assess whether the structural embeddings provide additional predictive power or improved biological interpretability for AMR prediction.

Three research papers in the area

1. <https://www.mdpi.com/2075-4418/15/3/279> - Demonstrates machine learning approaches for TB AMR prediction using genomic data using the same dataset provided for this project.

2. <https://www.nature.com/articles/s41586-024-07487-w> - AlphaFold 3 paper describing

capabilities for modelling protein complexes with nucleic acids

3. <https://academic.oup.com/bioinformatics/article/40/10/btae550/7754486> – paper showing utility in the use of protein structures for AMR prediction

Biological systems involved, Area of study, Other key words associated with the project

Prokaryote; Treatment/Therapeutics; Evolution; Genomic; Methods; Structural Biology

Genomics; Proteomics; Population genomics; Phylogenetics/phylogenomics; Machine; Learning/AI; Database searching

Bioinformatics field, Biological data & Programming language(s) used

DNA; Proteins; RNA; Python; R; Bash; AlphaFold 3

Helpful previous knowledge

Some experience with the command-line, python and HPCs, an understanding of differences between population genetics of eukaryotes and prokaryotes, the basics of protein structural biology, and antimicrobial resistance.

37. Increasing analysis throughput for an assay to model kidney health

Dr Becky Foster, Prof Simon Satchell and Dr Michael Crompton

becky.foster@bristol.ac.uk

General research field

Analysis throughput of a pre-clinical model for kidney disease

You would be working within Bristol Renal, a large research group in Bristol Medical School, run by a mixture of fundamental scientists and clinical academics

Project description

End stage renal disease is on the rise, particularly that caused by diabetes and hypertension, both of which are considered glomerular diseases. The glomerulus is the ball of blood vessels in the kidney that filter blood into urine. The mainstay of treatment for these conditions is to block the renin angiotensin system, although this approach is clearly not effective enough. Therefore, there is an urgent need for new clinical treatments that prevent glomerular disease. A suitable screening system is critical to enable us to identify appropriate treatments.

We have developed a screening system using glomeruli (balls of blood vessels) isolated from rat and mouse kidneys (1). This involves loading the glomeruli with green-labelled albumin and imaging the glomeruli using a confocal microscope. In brief, the glomerulus is held in place and sits in a bath of green labelled albumin. The bath is replaced with non-labelled albumin in a wash step and then the green labelled albumin inside the glomerulus diffuses out. We measure the rate of decline in signal in a single capillary loop. The loss of green labelled albumin from inside glomerular capillaries is captured using videos and analysed off-line, using region of interests (ROI) and applying an equation derived from Fick's Law of diffusion. The analysis is a bottleneck for throughput. We hypothesise that analysis can be automated, using AI and machine learning.

Video files will be converted to a TIFF series using Image J. Deep learning will be utilised on existing example image series (e.g. 50). ROI will be highlighted and areas not of interest will also be noted, including capillary loops containing red blood cells. The ROI can track the capillary loop where there is minor movement of the glomerulus during the wash stage. It is anticipated that capillary loop identification can be automated, appropriate ROIs placed automatically and Ps'alb calculated automatically. Same level of glomerular drift or warping can be accounted for within the software.

Three research papers in the area

(1). Desideri, S., Onions, K.L., Qiu, Y., Ramnath, R.D., Butler, M.J., Neal, C.R., King, M.L.R., Salmon, A.E., Saleem, M.A., Welsh, G.I., Michel, C.C., et al. (2018). A novel assay provides sensitive measurement of physiologically relevant changes in albumin permeability in isolated human and rodent glomeruli. *Kidney Int* 93, 1086-1097. 10.1016/j.kint.2017.12.003.

Biological systems involved, Area of study, Other key words associated with the project

Animal; Treatment/Therapeutics; Medical/clinical; Methods; AI, automated, machine learning, kidney disease, glomerulus, drug screening

Bioinformatics field, Biological data & Programming language(s) used

Machine Learning/AI; Images/video; Image J/Fiji; LAS (leica); Excel; Prism

Helpful previous knowledge

This team is currently using this assay with industry partners to help show mechanism of action of their drugs on glomeruli and to provide supportive information for FDA approval of drugs to be used in treating patients. This assay is sensitive and reproducible, but analysis from different experiments can take weeks. This project will have a real impact on all users of this assay, and will increase throughput of analysis. This is of particular importance to attract new industry partners in the future.

Other relevant information

You will be working alongside Dr Mike Crompton, one of the supervisory team and a senior research associate currently working on this project. The team has expertise in microvascular disease and glomerular physiology.

38. Developing the first GWAS associated with vascular glycocalyx changes in Avon longitudinal study of children of the 90's (ALSPAC)

Dr Becky Foster, Prof Simon Satchell, Prof Emma Vincent , Dr Joshua Burton

becky.foster@bristol.ac.uk

General research field

Population data analysis

Project description

The glycocalyx is a protective gel-like layer that lines the endothelial cells of all blood vessels and plays a crucial role in vascular health. Damage to this layer can cause early steps in vascular disease progression (1,2).

Glycocalyx measures in patients or the general population are very limited, which restricts meaningful research and therapeutic development. Indirect measures of systemic glycocalyx depth (termed perfused boundary region) (3) have been taken in the ALSPAC cohort (<http://www.bristol.ac.uk/alspac/researchers/our-data/>) using sidestream dark field imaging, in conjunction with Glycocheck software analysis. We have cleaned these data, which now present the potential for investigating genome-wide associated changes (data associated with the cohort) with perfused boundary region measures, creating a glycocalyx-related GWAS, for the first time.

Three research papers in the area

1. Gomez Toledo, A., Golden, G.J., Cummings, R.D., Malmström, J., and Esko, J.D. (2025). Endothelial Glycocalyx Turnover in Vascular Health and Disease: Rethinking Endothelial Dysfunction. *Annu Rev Biochem* 94, 561-586. 10.1146/annurev-biochem-032620-104745.
2. Gamez, M., Ramnath, R.D., Butler, M.J., Foster, R.R., and Satchell, S.C. (2025). The glomerular endothelial glycocalyx as a therapeutic target in proteinuric kidney disease. *Nature reviews. Nephrology*. 10.1038/s41581-025-01028-y.
3. Nieuwdorp, M., Meuwese, M.C., Mooij, H.L., Ince, C., Broekhuizen, L.N., Kastelein, J.J., Stroes, E.S., and Vink, H. (2008). Measuring endothelial glycocalyx dimensions in humans: a potential novel tool to monitor vascular vulnerability. *Journal of applied physiology (Bethesda, Md. : 1985)* 104, 845-852. 10.1152/jappphysiol.00440.2007.

Biological systems involved, Area of study, Other key words associated with the project

Human; Medical/clinical;Glycocalyx, ALSPAC, GWAS, perfused boundary region, vascular disease.

Bioinformatics field, Biological data & Programming language(s) used

Database searching;Population genomics; DNA; R;Bash

Helpful previous knowledge

Dr Foster and Prof Satchell have previously built knowledge on the importance of the glycocalyx layer in blood vessel barrier function, particularly in diabetes. They have used human cell models and animal models to prove that it can be targeted therapeutically to restore blood vessel function. Glycocalyx measures in large cohorts would address an unmet need to enable research to progress in this area towards clinical application.

Other relevant information

You will be working alongside Dr Josh Burton, one of the supervisory team and a senior research associate currently working on this project. The team has expertise in microvascular disease and glycocalyx biology (Foster and Satchell) and genetic epidemiology (Vincent and Burton).

39. Incidence of reoperation after primary hip or knee replacement in England

Dr Josh Lamb and Dr Adrian Sayers

josh.lamb@bristol.ac.uk

Other research staff, PhD students of technicians involved (if applicable)

NJR Lot 2 Contract team

General research field

Lower limb joint replacement surgery

Project description

Over 200 thousand hip or knee replacements are performed in England on a yearly basis and the outcomes of these procedures are used to evaluate quality of care, hospital, surgeon and implant performance. Currently revision operations (change of all or part of a joint replacement) is used as a surrogate for failure of the operation. A suitable student will aim to perform an analysis of English NHS data to understand what the incidence of any reoperation are after primary hip or knee replacement in the NHS. Secondary aims include the estimation of the types of reoperation. This information will inform the development of future outcome measures after primary hip or knee replacement.

Three research papers in the area

Jeschke E, Gehrke T, Günster C, Heller KD, Leicht H, Malzahn J, Niethard FU, Schröder P, Zacher J, Halder AM. Reoperation and Complication Rates after Hip and Knee Replacement Surgery in 1 046 145 Obese Patients. Dtsch Arztebl Int. 2023 Jul 24;120(29-30):501-502. doi: 10.3238/arztebl.m2023.0067. PMID: 37981818; PMCID: PMC10511005.

Causes of Reoperations After Primary Total Hip Arthroplasty: A Retrospective Cohort Study Over 20 Years

Park, Ki-Tae et al. The Journal of Arthroplasty, Volume 40, Issue 9, 2338 - 2346.e1

Lamb JN, Sayers A, Wilkinson JM, Pandit H, Whitehouse MR. The association between implant design, age, sex and the rate of major reoperation in patients undergoing primary total hip replacement: A retrospective study of UK National Joint Registry and Hospital Episodes Statistics data. PLoS Med. 2025 Nov 26;22(11):e1004538. doi: 10.1371/journal.pmed.1004538. PMID: 41296797; PMCID: PMC12680321.

Biological systems involved, Area of study, Other key words associated with the project

Human; Medical/clinical; Treatment/Therapeutics; Methods

Bioinformatics field, Biological data & Programming language(s) used

Data Modeling; Database searching; Hospital episode statistics; R; Python; Stata; R or Stata

Helpful previous knowledge

STATA or R programming skills are essential. Previous familiarity with hospital episode data or managing very large datasets is desired.

40. Identifying gene expression signatures for kidney disease phenotypes

Dr Amy Osborne

bb19705@bristol.ac.uk

Second project supervisor
Prof Moin Saleem

General research field

Project 1: genomics

Project description

Nephrotic syndrome (NS), a rare kidney disease, leads to the breakdown of the glomerular filtration barrier, leading to a loss of essential circulating proteins into the urine. NS is a heterogenous disease with different clinical classification criteria used in children and adults. Paediatric patients are grouped by their response to steroid treatment, whereas adult patients are classified by renal histology results. Defining novel NS subgroups by disease mechanism at the molecular level, including genomic and transcriptomic analyses, would help research develop more targeted, effective treatments for NS.

This project will involve the analysis of multiple published kidney disease transcriptomics datasets available in the Gene Expression Omnibus (GEO), for NS, using tools such as GEO2R (involving DSeq2 and volcano plots), to look for significantly differentially expressed genes across NS subtypes and/or compared to healthy controls.

Following identification of differentially expressed genes, a more detailed follow-up in NURTuRE and/or NephroS International datasets using machine learning or statistical methods would be employed.

Three research papers in the area

- Colby E, Hayward S, Benavente M, Robertson F, Bierzynska A, Osborne A, Parmesar K, Afzal M, Chapman T, Ullah F, Davies E, Nation M, Cook W, Johnson T, Andag U, Radresa O, Skroblin P, Bayerlova M, Unwin R, Vuilleumier N, Banks RE, Braddon F, Koziell A, Taal MW, Welsh GI, Saleem MA. National Unified Renal Translational Research Enterprise: Idiopathic Nephrotic Syndrome (NURTuRE-INS) study. Clin Kidney J. 2024 Mar 30;17(8):sfae096. doi: 10.1093/ckj/sfae096. PMID: 39135942; PMCID: PMC11317841.

- Bierzynska A, McCarthy HJ, Soderquest K et al. Genomic and clinical profiling of a national nephrotic syndrome cohort advocates a precision medicine approach to disease management. Kidney Int 2017;91:937–47. 10.1016/j.kint.2016.10.013

- Bezdzicka M, Cinek O, Semjonov V, Polackova K, Sladkova E, Zieg J, Saleem MA, Soucek O. Nephrotic syndrome sera induce different transcriptomes in podocytes based on the steroid response. Physiol Rep. 2024 Feb;12(3):e15932. doi: 10.14814/phy2.15932. PMID: 38307723; PMCID: PMC10837055.

Biological systems involved, Area of study, Other key words associated with the project

Human;Genomic;Medical/clinical;Exploratory

Bioinformatics field, Biological data & Programming language(s) used

Genomics;Transcriptomics;Population genomics;Machine Learning/AI;DNA;RNA;Python;R;Bash;

GEO2R (online), DESeq2 (R) or PyDESeq2 (Python), possibly other R or Python packages for transcriptomics data analysis.

41. Identifying multiple genetic variants associated with kidney disease subtypes using metaCCA and other joint statistical tests

Dr Amy Osborne

2 projects available

bb19705@bristol.ac.uk

Second project supervisor
Colin Campbell

Other research staff, PhD students of technicians involved (if applicable)
Prof Moin Saleem

General research field
transcriptomics

Project description

Nephrotic syndrome (NS), a rare kidney disease, leads to the breakdown of the glomerular filtration barrier, leading to a loss of essential circulating proteins into the urine. NS is a heterogeneous disease with different clinical classification criteria used in children and adults. Paediatric patients are grouped by their response to steroid treatment, whereas adult patients are classified by renal histology results. Defining novel NS subgroups by disease mechanism at the molecular level, including genomic and transcriptomic analyses, would help research develop more targeted, effective treatments for NS.

In genomic analyses, statistical methods such as linear regression based genome-wide association study (GWAS) and Polygenic Risk Scores (PRS) are used to identify common genetic risk variants, such as single nucleotide polymorphisms (SNPs), associated with phenotypes (such as disease status). However, the multivariate-SNP methods canonical correlation analysis (CCA) and metaCCA have not previously been used to look for multiple SNP sets associated with NS risk.

For this project, analysis of a published GWAS summary statistic SNP dataset (Barry et al., 2023) would be undertaken to look for associations of multiple SNP sets with the steroid sensitive (SS) nephrotic syndrome phenotype. This would involve running metaCCA 'multivariate-SNP' analyses (in R), or similar, to compute joint association tests of SNP sets with the NS case/control phenotype. The focus would be on SNPs either associated with (e.g. by published expression quantitative trait loci (eQTL), protein QTL (pQTL) studies), or located in, known genes (e.g. immune or kidney genes), and also include gene-level SNP aggregation tests. This project could also be extended to other kidney diseases for which there are published GWAS summary statistics datasets, and other joint SNP tests and potentially machine learning methods.

Three research papers in the area

- Osborne AJ, Bierzynska A, Colby E, Andag U, Kalra PA, Radresa O, Skroblin P, Taal MW, Welsh GI, Saleem MA, Campbell C. Multivariate canonical correlation analysis identifies additional genetic variants for chronic kidney disease. NPJ Syst Biol Appl. 2024 Mar 9;10(1):28. doi: 10.1038/s41540-024-00350-8. PMID: 38459044; PMCID: PMC10924093.
- Barry A, McNulty MT, Jia X, Gupta Y, Debiec H, Luo Y, Nagano C, Horinouchi T, Jung S, Colucci M, Ahram DF, Mitrotti A, Sinha A, Teeninga N, Jin G, Shril S, Caridi G, Bodria M, Lim TY, Westland R, Zanon F, Marasa M, Turudic D, Giordano M, Gesualdo L, Magistroni R, Pisani I, Fiaccadori E,

Reiterova J, Maringhini S, Morello W, Montini G, Weng PL, Scolari F, Saraga M, Tasic V, Santoro D, van Wijk JAE, Milošević D, Kawai Y, Kiryluk K, Pollak MR, Gharavi A, Lin F, Simões E Silva AC, Loos RJF, Kenny EE, Schreuder MF, Zúrowska A, Dossier C, Ariceta G, Drozowska-Duklas M, Hogan J, Jankauskiene A, Hildebrandt F, Prikhodina L, Song K, Bagga A, Cheong H 2nd, Ghiggeri GM, Vachvanichsanong P, Nozu K, Lee D, Vivarelli M, Raychaudhuri S, Tokunaga K, Sanna-Cherchi S, Ronco P, Iijima K, Sampson MG. Multi-population genome-wide association study implicates immune and non-immune factors in pediatric steroid-sensitive nephrotic syndrome. *Nat Commun*. 2023 Apr 29;14(1):2481. doi: 10.1038/s41467-023-37985-w. PMID: 37120605; PMCID: PMC10148875.

- Cichonska A, Rousu J, Marttinen P, Kangas AJ, Soininen P, Lehtimäki T, Raitakari OT, Järvelin MR, Salomaa V, Ala-Korpela M, Ripatti S, Pirinen M. metaCCA: summary statistics-based multivariate meta-analysis of genome-wide association studies using canonical correlation analysis. *Bioinformatics*. 2016 Jul 1;32(13):1981-9. doi: 10.1093/bioinformatics/btw052. Epub 2016 Feb 19. PMID: 27153689; PMCID: PMC4920109.

Biological systems involved, Area of study, Other key words associated with the project
Human;Genomic;Medical/clinical;Exploratory

Bioinformatics field, Biological data & Programming language(s) used
Genomics;Transcriptomics;Population genomics;Machine Learning/AI;DNA;RNA;Python;R;Bash;
metaCCA (R), Plink (bash)

42. Unravelling the Molecular Mechanisms of Skin Infection using Computational Approaches

Dr Rebecca Notman

rebecca.notman@bristol.ac.uk

General research field

Computational identification and analysis of protein–protein interactions

Project description

Skin diseases are a significant global health problem, affecting an estimated 20%-25% of the world's population at any given time, according to the World Health Organisation. Skin infections are one of the most common and burdensome types of skin disease, often leading to complications that impact quality of life and, in severe cases, can become life-threatening. As resistance to existing treatments increases, there is an urgent need to understand how these infections take hold in order to identify new ways to prevent and treat them.

Colonisation and infection of the skin rely on interactions between the surface proteins of pathogens and proteins in the uppermost layers of the skin. However, the molecular mechanisms that enable pathogens to bind to and disrupt skin proteins remain poorly understood. Investigating these interactions at the molecular level is crucial for identifying new strategies to prevent and treat infections. This MSc project offers an opportunity to explore how pathogens breach the skin's protective barrier using modern computational approaches.

The project will use a variety of computational techniques to identify and analyse pathogen-skin interactions, which will advance our understanding of the initiation and progression of skin infections at the molecular level. The project will begin with a systematic, sequence-based analysis of publicly available datasets to identify and prioritise candidate host-pathogen pairs. For a small number of high-priority candidates, structural information will be incorporated to generate hypotheses about plausible interaction modes between bacterial and skin proteins. Finally, molecular simulations will be used to explore the stability and behaviour of proposed protein-protein complexes and to refine the structural models. The insights gained will have application in designing new therapeutic strategies to enhance the skin's defences against infection.

Three research papers in the area

- (1) Foster, T. J. Surface Proteins of *Staphylococcus epidermidis*. *Frontiers in Microbiology* 2020, Volume 11 - 2020. DOI: 10.3389/fmicb.2020.01829.
- (2) Paiva, T. O.; Viljoen, A.; da Costa, T. M.; Geoghegan, J. A.; Dufrêne, Y. F. Interaction of the *Staphylococcus aureus* Surface Protein FnBPB with Corneodesmosin Involves Two Distinct, Extremely Strong Bonds. *ACS Nanoscience Au* 2023, 3 (1), 58–66. DOI: 10.1021/acsnanoscienceau.2c00036.
- (3) Henderson, S. R.; Geoghegan, J. A. The A domain of clonal complex 1-type fibronectin binding protein B promotes adherence and biofilm formation in *Staphylococcus aureus*. *Microbiology* 2023, 169 (6). DOI: <https://doi.org/10.1099/mic.0.001348>.

Biological systems involved, Area of study, Other key words associated with the project

Prokaryote;Eukaryote;Human;Medical/clinical;Structural Biology;Biochemistry;Protein-protein interactions; Biomolecular simulation

Bioinformatics field, Biological data & Programming language(s) used

Machine Learning/AI;Database searching;Proteins;Python;Bash;Gromacs for biomolecular simulations and VMD or PyMOL for visualisation.

Helpful previous knowledge

An understanding of protein structure and (bio)molecular interactions. Familiarity with command-line tools and the use of Python for data analysis. Familiarity with tools to identify and characterise protein sequences. Prior knowledge of molecular modelling and simulation is advantageous but not required as full training will be provided.

43. Open projects on evolution - Pisani Lab

Prof. Davide Pisani

3 projects available

Davide.Pisani@bristol.ac.uk

General research field

Evolutionary Biology

Project description

Projects on protein family evolution and organismal evolution and is happy to supervise students working across all types of cellular organisms. From prokaryotes to humans. But no viruses. The project will be discussed and decided in collaboration with the student based on their current interest in the lab.

Three research papers in the area

<https://www.nature.com/articles/s41559-023-02299-z> (the molecular clock analysis - in main paper; and the opsin analysis - is supplementary).

<https://www.nature.com/articles/s41559-024-02461-1>

<https://www.pnas.org/doi/full/10.1073/pnas.1204609109>

<https://www.biorxiv.org/content/10.1101/2024.06.24.600355v2>

Biological systems involved, Area of study, Other key words associated with the project

Prokaryote;Eukaryote;Human; Evolution

Bioinformatics field, Biological data & Programming language(s) used

Phylogenetics/phylogenomics;Database searching;Proteomics;Genomics; DNA;RNA;Proteins

R;Bash;Python;

44. Open projects on evolution, comparative genomics, molecular phylogenetics - Paps Montserrat lab

Jordi Paps Montserrat

4 projects available

jordi.paps@bristol.ac.uk

General research field

Evolution, comparative genomics, molecular phylogenetics

Project title

Open project on evolution, comparative genomics, molecular phylogenetics

Project description

The history of life on Earth shows major evolutionary transitions, shifts in which new biological features emerged that dramatically changed the biology of organisms. The endless forms most beautiful we observe today are the result of these revolutions that shaped the biology of the planet. They include the origins of animals, plants, and parasites among many others. Projects in our lab aim to unravel the molecular changes underlying these major transitions by analysing genomic data with comparative genomics and molecular phylogenetics. The reconstruction of ancestral genomes and the understanding of the evolution of key gene families are essential to comprehend the genesis of adaptations and biological innovations.

Three research papers in the area

JORDI PAPS & PETER W.H. HOLLAND (2018). Reconstruction of the ancestral metazoan genome reveals an increase in genomic novelty. Nature Communications (9):1730 JORDI PAPS, FEI XU, GUOFAN ZHANG & PETER W.H. HOLLAND (2015). Reinforcing the egg-timer: recruitment of novel lophotrochozoan homeobox genes to early and late development in the pacific oyster. Genome Biology and Evolution 7 (3): 677-688. JORDI PAPS, LUIS A. MEDINA-CHACÓN, WYTH MARSHALL, HIROSHI SUGA & IÑAKI RUIZ-TRILLO (2013). A molecular phylogeny of Apusozoa and Opisthokonts. Protist 164(1): 2–12

Biological systems involved, Area of study, Other key words associated with the project

Prokaryote;Eukaryote;Human; Evolution;Genomic;

Bioinformatics field, Biological data & Programming language(s) used

Comparative genomics; DNA;RNA;Proteins R;Bash;Python; MAFFT, IQTree, Phylobayes, Tree editors

Helpful previous knowledge

Evolutionary Biology

45. Open projects on engineering of biological systems - Gorochoowski lab

Prof Thomas Gorochoowski

4 Projects available

thomas.gorochoowski@bristol.ac.uk

General Field

Tools to aid in the engineering of biology

Project Description

I am happy to support projects that explore the development of tools to aid in the understanding or engineering of biological systems. This could involve the analysis of sequencing data sets of engineered cells to better understand the function of the synthetic genetic components introduced; the development of tools to help automate decision making when engineering genomes; methods to improve the standardised processing of data collected through diverse assays (e.g., sequencing, flow cytometry, etc.); and clever ways to visualise complex genetic circuitry.

Three research papers in the area

– Genetic circuit characterization and debugging using RNA-seq. Gorochoowski T.E., Espah Borujeni A., Park Y., Nielsen A.A.K., Zhang J., Der B.S., Gordon D.B. & Voigt C.A. Molecular Systems Biology 13, 952, 2017. (<http://msb.embopress.org/content/13/11/952>)

– Accelerated design of Escherichia coli genomes with reduced size using a whole-cell model and machine learning surrogate. Gherman I.M., Sharma K., Rees-Garbutt J., Pang W., Abdallah Z., Gorochoowski T.E., Grierson C.S. & Marucci L. Cell Systems 16, 101392, 2025. ([https://authors.elsevier.com/sd/article/S2405-4712\(25\)00225-X](https://authors.elsevier.com/sd/article/S2405-4712(25)00225-X))

– paraSBOLv: a foundation for standard-compliant genetic design visualization tools. Clark C.C., Scott-Brown J. & Gorochoowski T.E. Synthetic Biology 6, ysab022, 2021. (<https://doi.org/10.1093/synbio/ysab022>)

46. Using phylogenetics to inform avian influenza epidemiology and multiplex serological surveillance

Dr Amy Thomas (Population Health Sciences, Bristol Medical School)

Dr Bidisha Kundu (Population Health Sciences, Bristol Medical School)

amyc.thomas@bristol.ac.uk

General research field
Phylogenetics, virology

Project Description

Aims

1. Use AIV sequence data to establish past trends in AIV circulation in Europe
2. Construct phylogenetic trees for AIV to support development and interpretation of a multiplex AIV serological assay

Approach

Using publicly available data on GISAID, sequences from AIV isolated from European wild birds will be acquired and organised to

- i) Reconstruct past trends in AIV circulation in European wild birds, particularly Anseriformes (ducks, geese, swans), specifically, identifying dominant AIV subtypes and the timing/duration of their appearance
- ii) Within dominant subtypes, phylogenetic trees will be constructed to identify strains representative of antigenic clusters
- iii) Strains within current and proposed antigen panels will be evaluated in light of phylogenetic findings

Research environment and impact

This project will be embedded in a Wellcome Trust Early Career Fellowship. The student will have the opportunity to participate in collaborator meetings (APHA, UKHSA, Institut Pasteur, University of Oxford, Abbotsbury Swannery) and directly influence the real-time development and deployment of a novel bead-based serological assay for AIV surveillance in Anseriformes.

Three research papers in the area

1. Nguyen DT, Jang Y, Nguyen TD, Jones J, Shepard SS, Yang H, Gerloff N, Fabrizio T, Nguyen LV, Inui K, Yang G, Creanga A, Wang L, Mai DT, Thor S, Stevens J, To TL, Wentworth DE, Nguyen T, Pham DV, Bryant JE, Davis CT. 2017. Shifting Clade Distribution, Reassortment, and Emergence of New Subtypes of Highly Pathogenic Avian Influenza A(H5) Viruses Collected from Vietnamese Poultry from 2012 to 2015. *J Virol* 91:10.1128/jvi.01708-16. <https://doi.org/10.1128/jvi.01708-16>
2. Lee J, Had eld J, Black A, Sibley TR, Neher RA, Bedford T and Huddleston J (2023), Joint visualization of seasonal influenza serology and phylogeny to inform vaccine composition. *Front. Bioinform.* 3:1069487. doi: 10.3389/fbinf.2023.1069487
3. Maryam Hayati et al. Phylogenetic identification of influenza virus candidates for seasonal vaccines. *Sci. Adv.* 9, eabp9185 (2023). DOI:10.1126/sciadv.abp9185

Helpful previous knowledge

Avian influenza viruses (AIV) pose an ongoing threat to animal and human health. Recent events such as the H5N1 2.3.4.4b panzootic, leading to the interstate outbreak in US dairy cattle have highlighted AIVs capacity for rapid spread and cross-species transmission. Surveillance in wild birds is often biased toward detections of highly pathogenic viruses in birds found dead, limiting our understanding of true infection dynamics. Serology can overcome this by revealing patterns of exposure and immunity independent of clinical signs, but traditional assays such as HI, NI, and microneutralisation are labour-intensive, requiring live virus and high-containment facilities.

High-throughput multiplex bead-based assays offer a scalable alternative, enabling simultaneous quantification of antibodies to many HA and NA antigens in small sample volumes (1µl). A key challenge is selecting representative recombinant antigens, given the complex, cross-reactive antibody profiles generated by repeated infections.

Phylogenetic analysis can guide antigen selection by identifying sequence clusters within HA and NA subtypes that capture major patterns of viral diversity. Incorporating these phylogenetically informed antigens improves the ability of multiplex serosurveys to detect past exposure, characterise immunity, and support epidemiological inference for avian influenza.